

MERRYLAND HIGH SCHOOLS ENTEBBE

KIGUNGU AND KATABI

S5 TERM ONE 2020 HOLIDAY WORK

ECONOMICS

TOPIC 2: PRICE THEORY.

Sub-Topic1: Introduction to price Analysis.

Price theory: This is concerned with the study of prices and is regarded as the basis of economic theory.

It is concerned with the economic behaviour of individual consumers, producers and resource owners.

It explains the production, allocation and pricing of goods and services.

PRICE. This is the exchange value of a commodity in terms of money.

OR: The amount of money that has to be given up in order to obtain a good or service or a factor input.

MARKET: This is an arrangement that brings together buyers and sellers to transact business at a particular period of time.

A market is not restricted to an area but it takes place in different ways like on phone, telefaxing, Internet, etc

In the market, buyers and sellers must communicate together and in so doing, they influence the price.

A market has the following characteristics:

- There should be buyers and sellers who participate in the exchange of a commodity.

- There should be commodities to exchange
- There should be a medium of exchange agreed upon and acceptable to all participants' e.g. barter or monetary exchange.
- There should be a price at which commodities are exchanged.

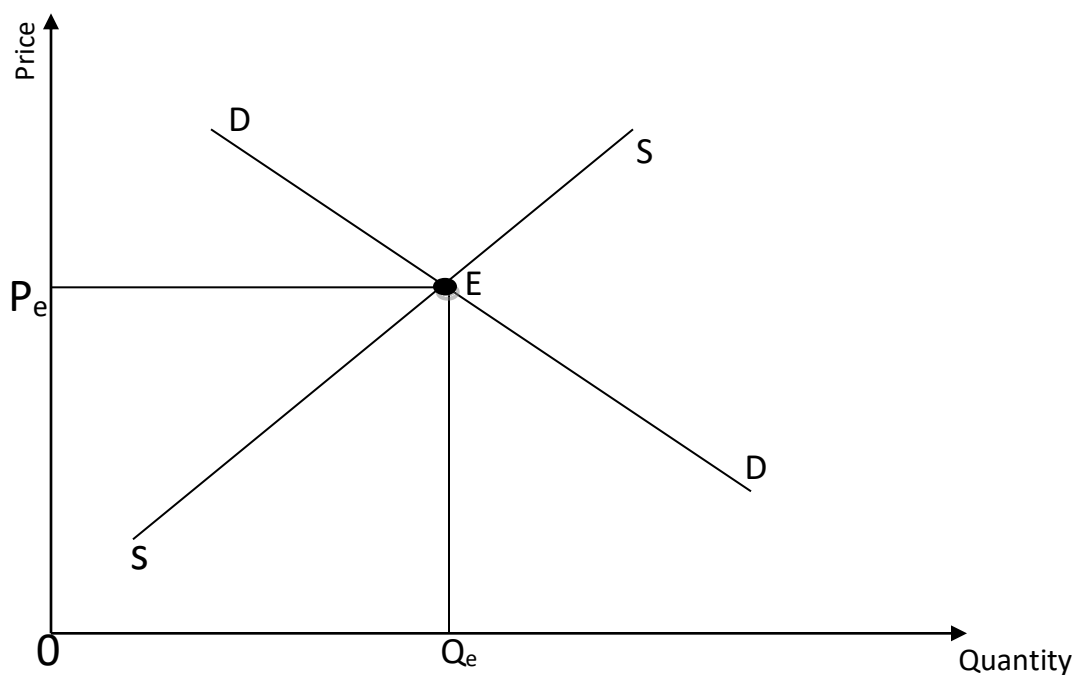
TYPES OF MARKETS:

1. **PRODUCT/ COMMODITY MARKETS;** these are markets in which goods or services are traded.
2. **RESOURCE/ FACTOR MARKETS;** these are markets in which production resources/factors of production especially labour and capital are traded.
3. **SPOT MARKETS;** these are markets where a commodity or a currency is traded for immediate delivery.
4. **FORWARD/ FUTURE MARKETS;** these are markets where buyers and sellers make a contract to buy or sell commodities at a fixed date at the price agreed upon in the contract/agreement.
5. **FREE MARKETS;** these are markets where government exerts no control/ intervention.
6. **CONTROLLED MARKET;** these are markets where the government or central authorities exerts a degree of control, for example by fixing prices, setting quotas etc.
7. **PERFECT MARKET;** this is the market where none of the buyers or sellers have the powers to influence prices in the market by either influencing demand or supply.
8. **IMPERFECT MARKETS;** this is where the buyer or seller has the power to influence the price in the market by either influencing demand or supply.
9. **ORGANISED MARKETS;** these are formal markets, such as a commodity market each dealing in a worldwide commodity, e.g. coffee, sugar, cocoa, rubber, etc.

TYPES OF PRICES

1. **NORMAL PRICE**; this is the one which is obtained where supply and demand are equal in the long run period.i.e. The long run equilibrium price.
2. **EQUILIBRIUM PRICE**; this is the price at which quantity supplied equals quantity demanded. It is determined by the interaction of the market forces of demand and supply. I.e. it is set or fixed at a point of intersection of demand and supply curves in a free enterprise economy.

Illustration of Equilibrium Price



From the above illustration **OP_e** is the equilibrium price and **OQ_e** is the equilibrium quantity and point **E** is the point of equilibrium

3. **RESERVE PRICE**; this is the price below which a seller is not willing to sell his/her product.

OR. It is the **least /lowest** possible acceptable price a seller can sell his or her product.

Determinants of Reserve Price

- Expectation of future demand for the product. Expectation of higher demand for the product in future leads to a high reserve price because the seller prefers to sell more in future and make more profits. On the other hand expectation of lower demand for the commodity in future leads to a lower reserve price because the seller prefers to sell currently and make more profits.
- **Level of durability or Perishability of the product.** Durable goods can be kept for a longer period of time and therefore a higher reserve price is fixed since the seller is not afraid of his/her product getting spoilt. On the other hand for perishable goods a lower reserve price is set because they cannot be kept for long period of time.
- **Cash flow requirements in the business.** The greater the need for cash in business the lower the reserve price set by the seller; this is because there is an urgent need for money in the business. On the other hand the less the need for cash in business the higher the reserve price set by the seller; this is so because there is no urgent need for cash in the business.
- **The storage costs in relation to future price.** The higher the storage costs, the lower the reserve price set by the seller this is so because the seller wants to sell off the products as fast as possible in order to reduce on the storage cost. On the other hand the lower the storage costs, the higher the reserve price set by the seller because the seller is not in a hurry to sell off his products since the storage costs are manageable
- **The length of time it takes before a new supply of goods reaches the market/.Gestation period.** The longer the period it takes for a new supply of goods to reach the market the higher the reserve price set by the seller; this is so because the seller is not scared of new supply of goods outcompeting the old stock. However the shorter the time it takes for new supply of goods to reach the market the lower the reserve price set by the seller since he/she wants to get rid of the old stock before the new stock reaches the market.

4. **MARKET PRICE**; this is the ruling/prevaling/reigning price of a product at a particular time.

OR: It refers to any price determined by buyers and sellers in the market in the short run period.

The market price may or may not necessarily be the equilibrium price since it is determined by a number of factors.

DETERMINANTS OF MARKET PRICE;

- **Through Haggling/Bargaining**; this is where a seller and a buyer carry out negotiations over the price until the two parties reach an agreeable price. Bargaining depends on the skills of a buyer and the seller. If the buyer has got more bargaining skills, then the price will be in his/her favour and if the seller has got more bargaining skills, then the price will be in his /her favour depending on the bargaining zone.
- **Through Auctioning/Bidding/tendering**. This is where a seller offers a product for sale and calls for bids (price offers) and the highest bidder takes the commodity. This is common in fund raising functions and sale of government property.
- **Through the market forces of demand and supply**; an increase in demand over supply leads to an increase in price on the other hand a fall in demand leads to a decrease in price.
- **Sale by treaties/agreements**; this is where buyers and sellers come together to fix the price of a given commodity e.g. the price of coffee is usually fixed by the international coffee agreement.
- **Price leadership**; this is where a dominant or low cost firm sets up a profit maximising price and other firms follow it.
- **Government policy of price legislations**; this is where the government fixes the price for the commodity through price control policy. It can either be a minimum price fixed above the equilibrium to protect the producers or fix a maximum price below the equilibrium to protect the consumers.

- **Offers at fixed price by individuals, government, institutions**, this is where a seller sets a price for his/her commodity and the buyer has to buy that commodity at that price e.g. prices in the supermarkets, government fixing wages of civil servants.
- **Collusion/cartel arrangements**; this is where different firms producing a similar commodity come together and agree on the price to charge for their product.
- **Resale price maintenance**; this is a practice where producers fix prices at which their products should be sold to the final consumer.

OR; this is a system where producers insist on fixing prices at which their products should be sold up to the retail level.

This is commonly used in the newspapers industry where manufactures fix prices at which consumers should buy these commodities.

MERITS OF RESALE PRICE MAINTENANCE;

- It helps to protect consumers from exploitation by middlemen.
- It helps to reduce competition especially between small scale and large scale retailers.
- Helps to maintain price stability.
- Business profits are easy to compute.
- Helps the seller to increase his profits through increased sales.
- It saves time since there is no need for bargaining.

FUNCTIONS OF PRICE IN THE MARKET:

- **Measuring the value of commodities**; the worth of commodities is expressed in terms of money. Guiding producers on what to produce. Producers normally go for commodities which fetch high prices.
- **Guiding consumers in making consumption decisions/plans**. Consumers put into consideration the prices attached to the various needs/wants before buying goods and services.
- **Determining income distribution** i.e. producers who sell their goods at higher prices earn more income than those who sell at low prices.
- **Guiding producers on how to produce/determining the technique of production to use** i.e. producers normally go for an affordable method of production in order to minimise the cost of production so as to maximise profits.
- **Guiding producers on where to produce/ choosing the best location for the business**. Producers always set up firms in those areas which have attractive markets and where consumers can afford to pay high prices for their goods so as to maximise profits.
- **Guiding the producer on deciding for whom to produce/ providing automatic adjustments between demand and supply**.

SUB-TOPIC2: THE THEORY OF DEMAND

DEMAND: This is defined as the desire for a commodity backed by the ability to pay a certain sum of money at a given price and time

OR: Demand is the quantity of goods which the consumers are willing to buy at a given price over a given period of time.

EFFECTIVE DEMAND: This is the actual buying of goods and services at a given price and at a given time.

OR. It is the actual amount of goods and services purchased by the consumer at a given price and at a given time

REASONS WHY PEOPLE DEMAND FOR GOODS:

- i. **FUNCTIONAL EFFECT;** some people buy certain goods because of the purpose they serve e.g. food for eating.
- ii. **VEBLEN/EXCLUSIVE CONSUMPTION;** one buys a commodity because he/she wants to be the only person identified with it i.e. the desire to be unique.
- iii. **SNOB EFFECT/CONSPICUOUS CONSUMPTION;** this is where an individual buys goods which are expensive just to show his economic power or status e.g. buying expensive designer clothes, expensive vehicle, unique phones etc.
- iv. **BANDWAGON EFFECT;** this is when a person buys a commodity because there are others buying it i.e. one buys a commodity in order to emulate others who have already bought it.
- v. **IMPULSIVE BUYING;** this is where an individual buys a commodity because he/she has seen it displayed i.e. the good is bought out of a sudden desire because the good is attractively displayed

TYPES OF DEMAND:

1. **Composite demand;** this is the total demand for a good with many uses/which can be used for more than one purpose. Examples of composite demand include;
 - Demand for electricity used for ironing, lighting, cooking.
 - The demand for wool for cloth making, cushioning, cleaning etc.
 - The demand for sugar for baking, sweetening drinks, brewing etc.
 - Demand for Iron and steel for construction, furniture making, manufacturing etc
2. **Joint/complementary demand;** this is demand for commodities that are used together in the satisfaction of human wants i.e. the buying of one commodity necessitates the buying of the other e.g. demand for a gun and bullets, demand for a car and fuel etc. Therefore the fall in the price of one commodity increases the quantity demanded of its complement and an increase in the price of one commodity leads to a fall in the quantity demanded of its complement.
3. **Competitive demand;** this is the demand for commodities that are close substitutes which serve **almost** the same purpose e.g. demand for butter and blue band, the demand for tea and coffee which are substitutes to each other.

A fall in price of one commodity reduces the demand for another (its substitute).

An increase in price of one commodity increases demand for another.(its substitute)

4. **Derived demand;** this refers to the demand for a commodity due to the demand for what it helps to produce e.g. demand for factors of production.
5. **Independent/autonomous demand;** this refers to demand for a commodity which has no effect or relationship with the demand for other commodities.

The demand function: This is a statement which shows a technical relationship between quantity demanded of a commodity and factors which influence it, such as price of a commodity (p), level of consumer's income (Y), prices of related commodities (p_r), tastes and preferences (T_p) etc.

i.e. $Q_d = f(P, Y, P_r, T_p \dots \dots \dots n)$.

DEMAND SCHEDULE: This is a table showing the amount of a commodity which is demanded by a consumer at different price levels.

The demand schedule reflects the law of demand which states that "the higher the price, the lower the quantity demanded and the lower the price the higher the quantity demanded holding other factors constant".

The demand schedule can be constructed for an individual or a group of individuals in the market.

An individual's demand schedule

Price (shs)	Quantity in kg
2000	5
1500	10
1000	15
500	20

The information on an individual demand schedule can be illustrated on the graph in order to come up with the demand curve

The demand curve: This is a graphical representation of quantity demanded of a commodity at different price levels.

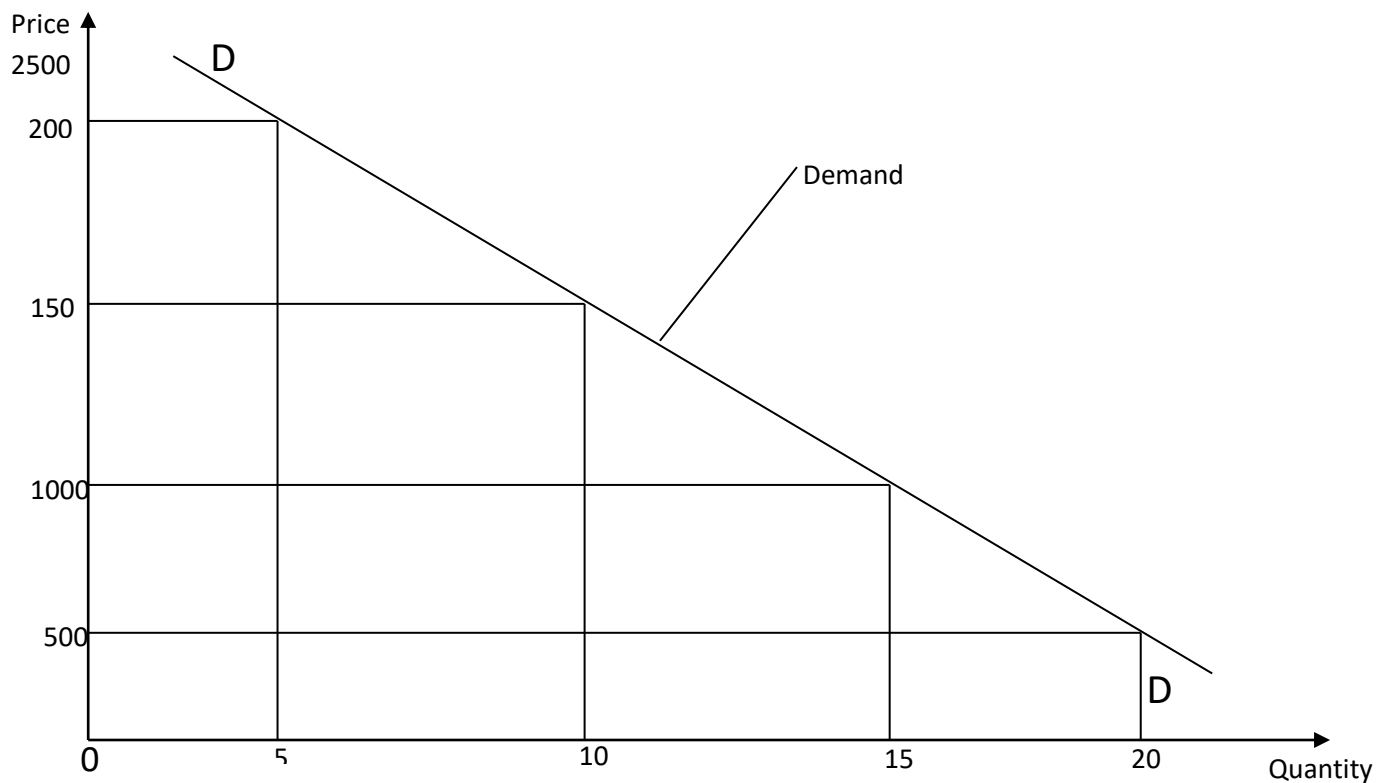
OR: It is a curve that shows quantity demanded of a commodity at different price levels.

Note: Price is represented on the vertical axis while quantity demanded on the horizontal axis.

A typical/normal demand curve is down ward sloping from left to right.

It is drawn on the assumption that quantity demanded depends on the price of the commodity, other factors affecting demand remaining constant

An illustration of the demand curve



From the diagram above, it is noted that at a higher price (Shs. 2000), 5kg are bought and as the price reduces e.g. to (Shs. 500) quantity demanded increases to 20kg.

3. Market demand. Market demand is the total demand of all the consumers of a given product at alternative prices in a given period of time.

If we sum up the different quantities of a commodity demanded by a number of individuals at various prices, we have a market demand schedule as shown below.

Price (Ug.Shs)	Demand of consumers				Market demand
	A	B	C	D	
60	6	3	1	0	10
50	10	5	3	2	20
40	14	8	5	3	30
30	16	12	7	5	40

THE LAW OF DEMAND: The law of demand states that the higher the price, the lower the quantity demanded of a commodity and the lower the price, the higher the quantity demanded of a commodity other factors affecting demand remaining constant/*Ceteris paribus*.

REASONS WHY THE DEMAND CURVE SLOPES DOWNWARDS FROM LEFT TO RIGHT OR REASONS WHY PEOPLE DEMAND MORE AT LOWER PRICES OR FACTORS THAT EXPLAIN THE LAW OF DEMAND:

1. Substitution effect. As the price of a commodity increases while prices of substitutes are constant, a commodity becomes relatively expensive in relation to its substitutes. Consumers therefore buy less of a commodity as they demand more of its substitutes which are relatively cheaper. However, as the price of a commodity decreases while prices of its substitutes remain constant, a commodity becomes relatively cheaper hence an increase in demand for it, thus leading to the downward sloping of the demand curve.

2. Real income effect. A fall in price leads to an increase consumer's real income. This means that the consumer can buy more units of a commodity using the same amount of income. However, as the price of a commodity increases the real income of a consumer falls hence less of a commodity is demanded.

3. The law of diminishing marginal utility. According to this law when one consumes more and more units of a commodity, the satisfaction he gets from each additional unit consumed diminishes/decreases. Therefore the consumer is only willing and ready to buy those extra units only when the price is reduced. This means that the consumer is willing to pay high prices for the first units of the commodity since they give higher satisfaction and pay less for the extra units to be consumed because of less satisfaction hence the downward sloping of the demand curve.

4. The price effect. A reduction in price of commodity it brings in more consumers and as a result demand increases while with an increase in price of a good, many consumers abandon that good which reduces consumption and decreases demand.

5. Different uses of a commodity. For a commodity that has many uses, an increase in price makes consumers use it for only vital purposes hence a decrease in demand. However, when the price decreases a commodity is put to various uses and its demand increases. For example, with the increase in the electricity tariffs, power is used primarily for domestic lighting, but when the tariffs are reduced, consumers use power for cooking, ironing, fans, and heaters.

6. Behaviour of low income consumers. The low income earners buy more of a commodity when the price reduces because they now afford it than when the commodity's price has increased and they cannot afford it hence the downward sloping of the demand curve.

ABNORMAL/EXCEPTIONAL DEMAND CURVES: These are curves that do not conform to the law of demand. Such curves do not slope down wards from left to right because more of a commodity may be demanded at a higher price or less of a commodity may be demanded at a lower price.

The cases of abnormal/exceptional demand:

- i. **In case of goods of ostentation (snob value goods);** these are goods consumed by the people as objects of pride/pomp. They are regarded as status symbols and are basically bought to impress others and therefore consumers prefer to buy them at higher prices rather than at lower prices.
- ii. **In case of giffen goods;** these are normally basic commodities which are consumed by the low income earners and their demand increases when their price rise. This is because these goods consume/ take up a large proportion of the consumer's income when their prices rise since the consumer can no longer afford alternative goods, i.e. the consumers abandon all other goods and concentrate on giffen goods.
- iii. **In case of expectation of price changes;** if the price of a commodity increases and there is an expectation of further increase in price, consumers increase the demand of a commodity in order to avoid buying the commodity at an even much higher price in future, similarly when the price of the commodity decreases and there is an expectation of further decrease in price consumers buy less so that they can buy a commodity in future at an even much lower price.

- iv. **In case of an effect of an economic depression;** an economic depression is a period of low economic activities and during this period, prices are low and demand for the goods is correspondingly low because people have low income.
- v. **In case of ignorance effect;** some consumers may mistake commodities of high prices to be of high quality which leads to greater quantities of a commodity being demanded at higher prices, this may be because of different packaging, designing, labelling etc.

THE FACTORS THAT AFFECT/INFLUENCE /DETERMIN THE DEMAND FOR A GIVEN COMMODITY (DETERMINANTS OF QUANTITY DEMANDED):

1. **Price of a commodity;** a high price of a commodity leads to low quantity demanded of such a commodity because consumers find it expensive to buy, while low price leads to high quantity demanded of a commodity because consumers find it cheap to buy.
2. **The price of a substitute good.** High price of a substitute good leads to high demand of the commodity in question; this is so because consumers find it cheaper to buy that commodity in question. However a low price of a substitute good leads to low demand for the commodity in question since consumers find it expensive to buy the commodity in question.
3. **The price of complement/complementary good;** High price of a complementary good leads to low demand for the commodity in question, this is so because less of the complementary good is bought which leads to low demand for the commodity in question since the two are used together in the satisfaction of human wants. However low price of a complementary good leads to high demand for a commodity in question, this is so because high quantity of a complementary good is bought since the two commodities are used together in the satisfaction of human wants.
4. **The level of the consumer income;** high level of consumer's level of income leads to high quantity demanded of a given commodity because the consumer has the capacity/ability to purchase the commodity. However low income of a consumer leads to low quantity demanded of a given commodity because of the low purchasing power of the consumer.
5. **Consumer's tastes and preferences;** favourable tastes and preferences lead to high demand for goods because of high interest for the commodity while unfavourable tastes and preferences leads to low demand for commodity since consumers have low interest for the commodity.
6. **The population size.** A high population size leads to high market for goods thus leading to high demand for such goods because there many potential buyers for such goods. On the other hand a low population size leads to low demand for goods because there a few potential buyers for such goods.
7. **Government policy as regards taxation or subsidisation of a commodity;** High level of taxation of a commodity leads to low demand for a commodity; this is so because high taxation makes the commodity expensive due to high price. On the other hand low level

of taxation of the commodity leads to high demand for a commodity, this is so because low taxation makes the commodity cheap due to low price.

8. **The nature of income distribution;** even distribution of income leads to high demand for goods, this is so because of the reasonable purchasing power of the majority of the people, while uneven distribution of income leads to low demand for goods because of the low purchasing power of the majority of the people.
9. **Seasonal factors;** Favourable season for a given commodity leads to high demand for it because there is apparent need. On the other hand unfavourable season/end of season leads to low demand for a good, this is so because there is no/limited apparent need.
10. **The degree of advertising;** A high degree of intensive and persuasive advertising lead to high demand for a good because of a high level of awareness by the consumers and being convinced to buy the good. On the other hand a low degree of advertising leads to low demand for a good, this is so because many consumers are not made aware of the existence of the good and convinced to buy.
11. **Availability of credit facilities.** Accessibility to credit facilities such as hire purchase leads to high demand for a commodity. This is so because the consumer continues acquiring the commodity even without ready cash, On the other hand absence of credit facilities such as hire purchase leads to low demand for a commodity, this is so because the consumer is unable to acquire the commodity without ready cash.

Factors that lead to high demand for a commodity:

- Low price of a commodity
- High price of a substitute good
- Low price of a complementary good
- High level of consumer's income
- High population size
- Favourable tastes and preferences
- Favourable season/Beginning of a season
- High level of advertising
- Even distribution of income among the population
- Low level of taxation of a commodity
- High accessibility to credit facilities.

Factors that lead to low demand for a commodity:

- High price of a commodity
- Low price of a substitute good
- High price of a complementary good
- Low level of consumer's income
- Low/small population size
- Unfavourable tastes and preferences
- Unfavourable season/ End of a season
- Low level of advertising
- Uneven distribution of income

- High level of taxation of a commodity
- Low accessibility to credit facilities.

WHY MAY CONSUMERS BUY LESS OF A COMMODITY WHEN IT'S PRICE FALLS?

- When the commodity is a giffen good.
- When there is anticipated further price reduction in future
- When consumers prefer goods of ostentation./Snob effect.
- When a fall in price is associated with a fall in quality.
- During a period of economic depression

WHY MAY CONSUMERS BUY MORE OF A COMMODITY WHEN IT'S PRICE INCREASES?

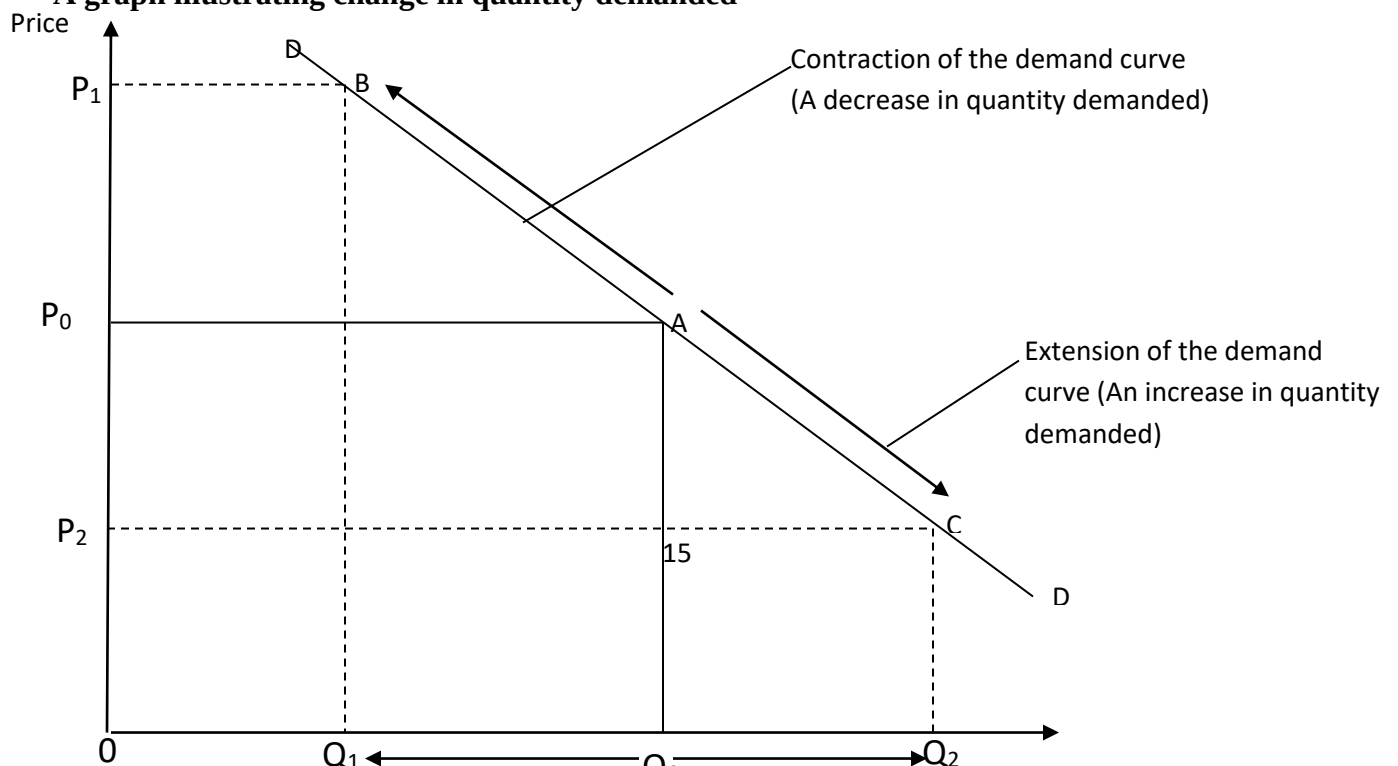
- When the commodity is a good of ostentation.
- When there is anticipated further increase in price in the future.
- In case of ignorance effect
- When there is persuasive advertising.
- During a period of economic prosperity.

CHANGE IN QUANTITY DEMANDED

This refers to an increase or decrease in the amount of the commodity demanded due to changes in its price, other factors that affect demand remaining constant.

It involves movement along the same demand curve either upwards or downwards.

A graph illustrating change in quantity demanded



Extension of the demand curve (an increase in quantity demanded)

This refers to a situation when more of a commodity is demanded due to a decrease in its, other factors that affect demand remaining constant. E.g. in the graph above the reduction in price from OP_0 to OP_2 leads to an increase in quantity demanded from OQ_0 to OQ_2

This is indicated by the downward movement along the same demand curve(A to C) as shown in the illustration above.

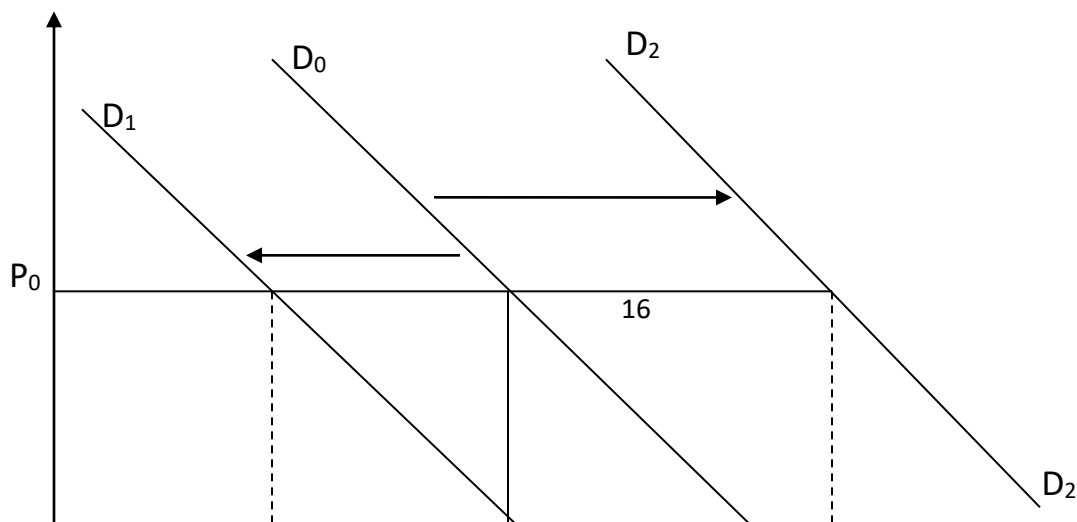
Contraction of the demand curve (a decrease in quantity demanded): This refers to a situation when less of a commodity is demanded due to **an increase in its price**, other factors that affect demand remaining constant.

E.g. in the graph above an increase in price from OP_0 to OP_1 leads to a decrease in quantity demanded from OQ_0 to OQ_1 .

This is indicated by an upward **movement along** the same demand curve i.e.(A to B)

CHANGE IN DEMAND: This refers to an increase or decrease in demand due to changes in other factors that affect demand price remaining constant. It involves a **shift** of the entire demand curve either to the right or to the left.

A graph illustrating a change in demand



In the graph above, at each possible price e.g. OP_0 , quantity demanded can increase or decrease due to changes in other factors that affect demand.

An increase in demand is illustrated by the shift of the demand curve to the right i.e. D_0 to D_2 , quantity demanded increases from OQ_0 to OQ_2 at a constant price OP_0 .

A decrease in demand is illustrated by the shift of the demand curve to the left. i.e. D_0 to D_1 . Quantity demanded decreases from OQ_0 to OQ_1 .

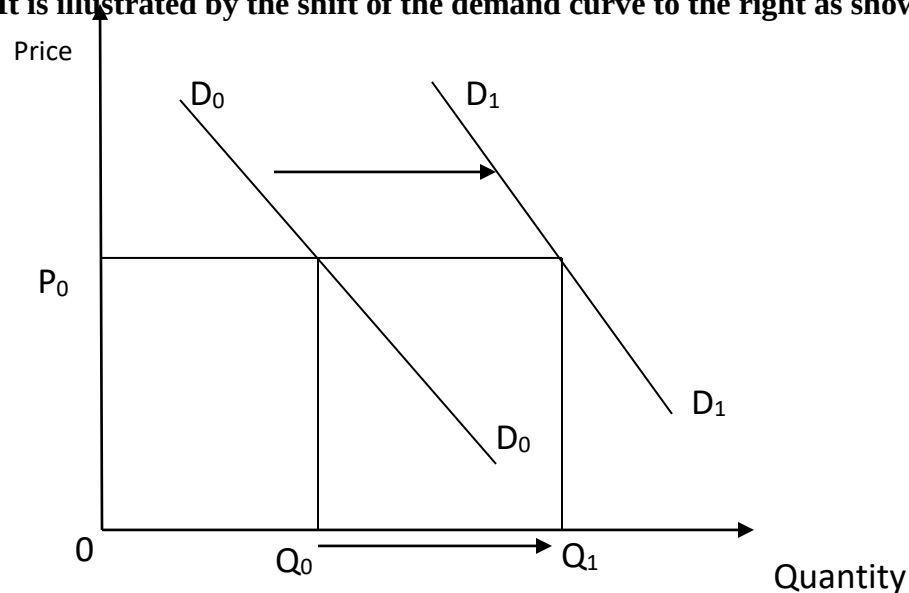
FACTORS THAT CAUSE A CHANGE IN DEMAND FOR A COMMODITY

- **A change in price of substitute commodity;** An increase in the price of a substitute good leads to an increase in demand for the commodity in question; this is so because consumers find it cheaper to buy that commodity in question. On the other hand a decrease in the price of a substitute good leads to a decrease in demand for the commodity in question, this is so because consumers find it expensive to buy the commodity in question
- **A change in the price of a complementary good;** an increase in price of a complementary good leads to a decrease in demand for the commodity in question, this is so because less of the complementary good is bought which leads to a decrease in demand for the commodity in question since the two are used together in the satisfaction of human wants. However a decrease in price of a complementary good leads to an increase in demand in its demand which leads to an increase in the demand for the commodity in question since the two are used together in satisfaction of human wants.

- **Change in the levels of income of the consumer;** An increase in the consumer's level of income leads to an increase quantity demanded of a given commodity because the consumer's increased capacity/ability to purchase the commodity. However a decrease in the income of a consumer leads to decrease in quantity demanded of a given commodity because of the reduced purchasing power of the consumer.
- **A change in consumer's tastes and preferences;** A favourable change in tastes and preferences lead to an increase in demand for a good because more people pick interest in the commodity while unfavourable change in tastes and preferences leads to a decrease in demand for it because many people lose interest in the commodity.
- **A change in population size;** An increase in population size leads to an increase in the market for a good thus leading to an increase in demand for such a good because there more potential buyers for such a good. On the other hand a decrease in population size leads to decrease in demand for a good because there are fewer potential buyers for such a good.
- **A change in the level of taxation on people's income;** An increased level of taxation on peoples income leads to a decrease in demand of a commodity due to the reduced purchasing power. However a decrease in the level of taxation on people's income leads to an increase in demand of a commodity due to the increased purchasing power of the consumer.
- **Changes in income distribution among households;** Income distribution becoming among households becoming even leads to an increase in demand for a good because it increases the purchasing power of the majority of the people, On the other hand income distribution among households becoming an even leads to a decrease in demand for good because it reduces the purchasing power of the majority of the people.
- **. Expectation of future changes in the price of the commodity;** an expectation of an increase in price of the commodity in future leads to an increase in demand for it, this is so because consumers buy more quantity of it currently so as to avoid buying it at an increased price. On the other hand expectation of a decrease in price of a commodity in future leads to a decrease in demand for it, this is so because consumers buy less quantity of it currently so as to buy more in future at a reduced price.
- **A change in seasons;** Favourable change in season for a given commodity leads to an increase demand for that commodity because there is increased need for it. On the other hand unfavourable change in season/ season coming to an end leads to a decrease in demand for a good, this is so because there is reduced need for it.
- **Change in the level of advertising;** an increased level of advertising leads to an increase in demand for a good because of the increased level of awareness by the consumers and being convinced to buy the good. On the other hand a decrease in the level of advertising leads to a decrease in demand for a good, this is so because of the reduced level of awareness among the consumers.
- **Change in availability of credit facilities;** increased accessibility to credit facilities leads to an increase in demand of a given commodity, this is so because it enables the consumer to access the good without ready cash. On the other hand reduced accessibility to credit facilities leads to a decrease in demand for a commodity, this is so because it becomes more difficult for consumers to access commodities without ready cash.

AN INCREASE IN DEMAND: This is a situation where more of the commodity is demanded due to **positive/favourable changes** in other factors that affect demand when its price is constant.

It is illustrated by the shift of the demand curve to the right as shown below.



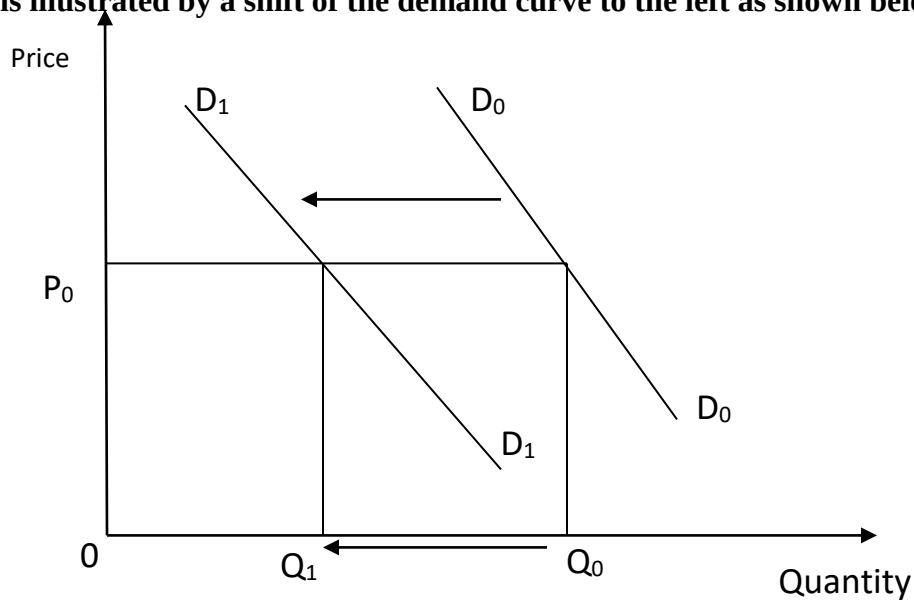
A shift of the demand curve from D_0D_0 to D_1D_1 indicates an increase in demand of a commodity from Q_0 to Q_1 at constant price OP_0

Causes of an increase in demand of commodity

1. An increase in the level of advertising.
2. An increase in price of substitute a good
3. An increase in consumer's income.
4. Favourable change in tastes and preferences/Tastes and preferences becoming favourable.
5. An increase in the population size.
6. A reduction in the level of taxation on people's income.
7. Increased access to credit facilities
8. A decrease in the price of a complementary good
9. Income distribution among households becoming fairer
10. Favourable change in season for a commodity
11. An expectation of a future increase in price of the commodity.

A DECREASE IN DEMAND: This is a situation where less of the commodity is demanded due to **unfavourable changes** in other factors that affect demand when price is constant.

It is illustrated by a shift of the demand curve to the left as shown below.



A shift of the demand curve from D_0D_0 to D_1D_1 illustrates a decrease in demand for a commodity from Q_0 to Q_1 at constant price OP_0 .

Causes of a decrease in demand for a commodity

1. A decrease in the population size.
2. A decrease in the level of advertising
3. An increase in the price of a complement
4. A decrease in price of a substitute
5. Season becoming unfavourable
6. Decrease in consumer's level income
7. Tastes and preferences becoming unfavourable/Unfavourable change in tastes and preferences
8. A decrease in access to credit facilities.
9. An increased level of taxation on people's income.
10. Income distribution among households becoming unfair/uneven
11. An expectation of a future decrease in the price of a commodity

THE THEORY OF CONSUMER BEHAVIOUR

A consumer is an individual, a household or government who uses goods and services to satisfy wants.

A consumer is always faced with a problem of allocating a fixed income among a variety of available options.

A consumer is assumed to be rational i.e. given his income and the market prices of the various Commodities; he plans the spending of his income so as to attain the highest possible utility

Basic approach to consumer behaviour

Consumer behaviour can explained using approaches the Cardinal utility theory/approach

. CARDINAL UTILITY THEORY/ APPROACH:

This is the utility where the satisfaction derived by the consumers from the consumption of a good or a service can be expressed numerically e.g. 1, 2, 3 etc.

NB: Utility is the satisfaction or pleasure derived from consumption of goods and services. It is assumed that, a consumer can know exactly how much satisfaction is derived from consumption of a good and such satisfaction is measured subjectively in units known as utils

Assumptions of cardinal utility theory/ approach:

- It assumes that a consumer is a rational being who calculates and measures, chooses and compares utilities of different units of goods thus maximizing utility.
- It assumes that a consumer possesses perfect knowledge of the choices open to him.
- It assumes that all commodities available to a consumer are perfectly divisible into smaller units.
- It assumes that as more a single commodity is consumed, total utility increases, reaches maximum level and then falls.
- It assumes that there are no perfect substitutes, and that utilities are measurable in terms of money.

Categories of utility

1. **Total utility.** This is the total satisfaction derived from consuming all different units of a given commodity in a particular period of time. For example, when a consumer buys apples, he receives them in units of 1, 2, 3, and 4. Two apples have more utility than one apple, three apples have more utility than two apples, and four apples have more utility than three apples.

2. **Marginal utility.** This is the additional satisfaction derived from consumption of an extra unit of a commodity in a particular period of time. For example, the total utility of two apples is 35 utils, and when a consumer consumes the third apple total utility becomes 45 utils. Therefore, the marginal utility of the third apple is equal to $(45-35 = 10 \text{ utils})$.

Marginal utility is given as:

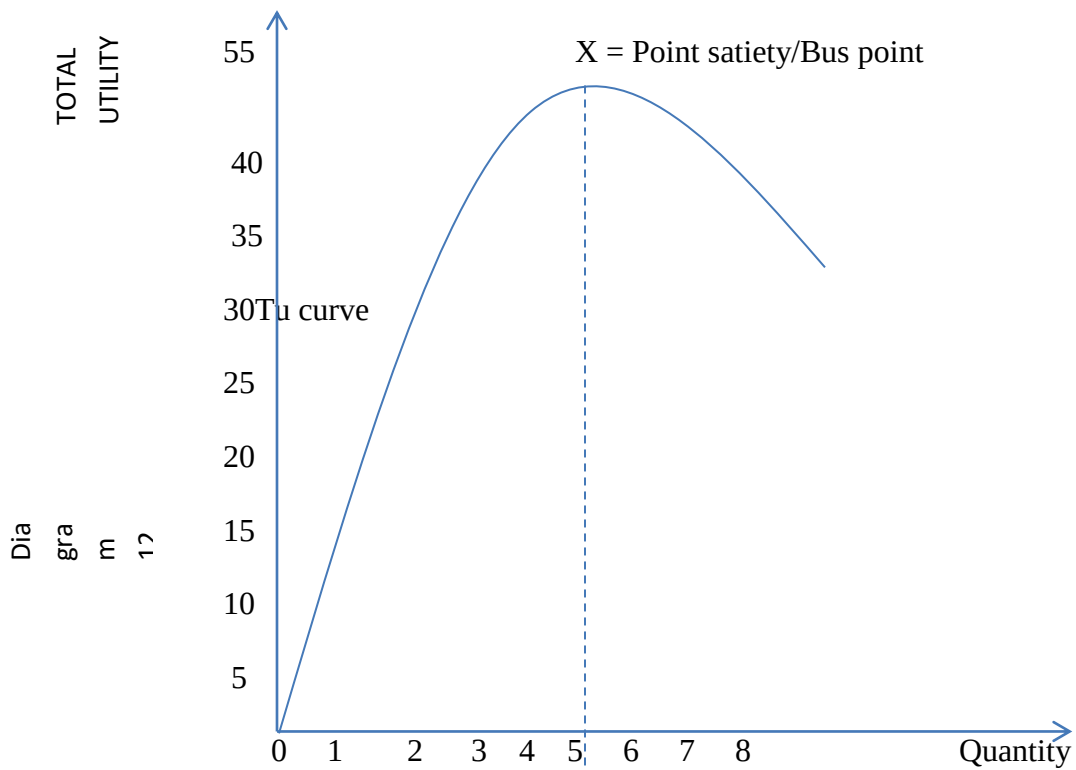
$$\text{Marginal utility} = \frac{\text{Change in total utility}}{\text{Change in quantity of a good}}$$

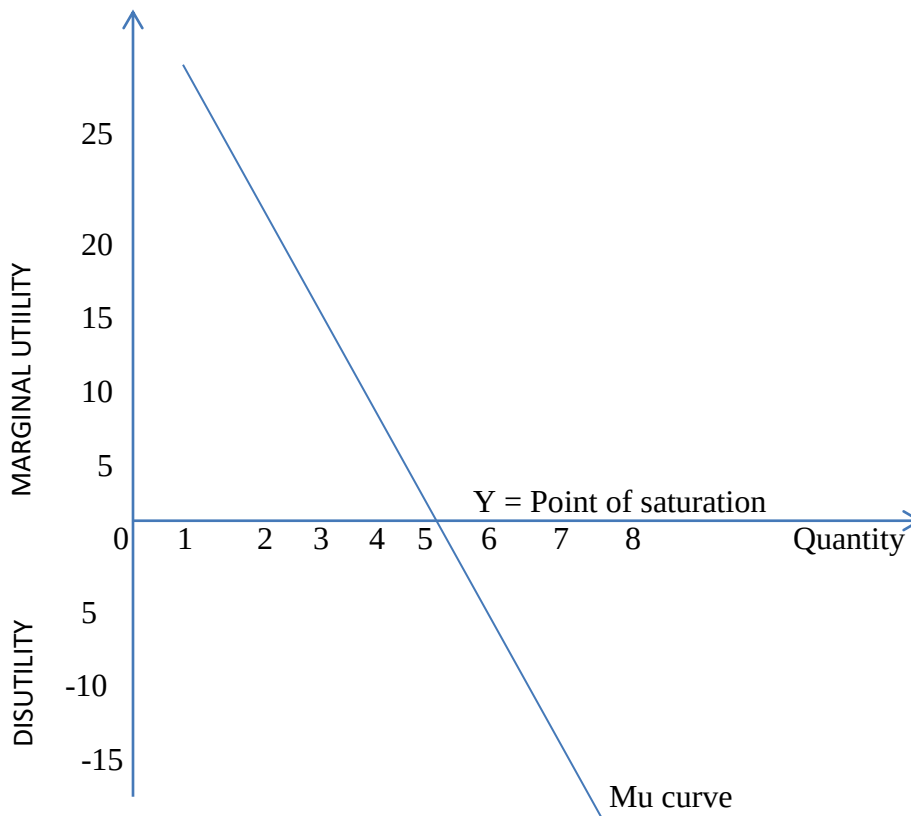
Table showing the relationship between total utility and marginal utility

Units of apples	Total utility	Marginal utility
0	0	-
1	20	20
2	35	15
3	45	10
4	50	5
5	50	0
6	45	-5
7	35	-10

The units of apples which a consumer chooses are in descending order of their utilities. The first apple is the best out of the lot available to him and hence gives him the highest satisfaction measured as 20 utils.. The second apple is naturally the second best with lesser amount of utility than the first and has 15 utils..

Diagram showing the relationship between total utility and marginal utility





- i. When total utility is increasing, marginal utility is decreasing
- ii. When total utility is at maximum i.e. point X (satiety/bliss point), marginal utility is at zero i.e. point Y.(point of saturation)
- iii. When total utility starts falling but still positive, marginal utility goes into negative (disutility)

THE LAW OF DIMINISHING MARGINAL UTILITY: The law of diminishing marginal utility states that as more and more units of a commodity are consumed, the satisfaction derived from each additional unit diminishes.

ASSUMPTIONS OF THE LAW OF DIMINISHING MARGINAL UTILITY:

- It assumes that there should be a single commodity with homogeneous units wanted by an individual consumer.
- It assumes that there should be continuity in consumption of a commodity i.e. units of a commodity should be consumed in succession at a particular time.
- It assumes that there should be no change in taste, habit and income of a consumer. A change in any of the mentioned factors is likely to increase rather than diminish utility
- .It assumes that prices of different units of a commodity should remain constant.
- It assumes that a consumer should be rational i.e. with a calculating mind, aims at maximizing utility.
- It assumes that all units of a commodity are of a standard size e.g. a sizeable glass to quench thirst and not a spoon.

- It assumes that a commodity should be divisible to allow successive consumption of a commodity.
- It assumes that the commodity should be of an ordinary type i.e. not superior good or goods of ostentation or addictive goods.

IMPORTANCE OF THE LAW OF DIMINISHING MARGINAL UTILITY

- It explains the phenomenon in the value theory that price of a commodity falls when its supply increases, because with increase in stock of a commodity marginal utility diminishes.
- The principle of progressive of taxation is based on this law. As a person's income increases, his/her rate of tax rises because marginal utility of money to him/her falls with rise in income.
- It explains the diamond-water paradox of Smith. Because of its scarcity, diamond possesses a high marginal utility and therefore commands high price, since water is relatively abundant, its marginal utility is low, it commands low price yet is more useful than diamond.
- It helps in bringing variety in production and consumption since consumption of the same good brings about boredom, hence its utility diminishes thus the desire for variety.

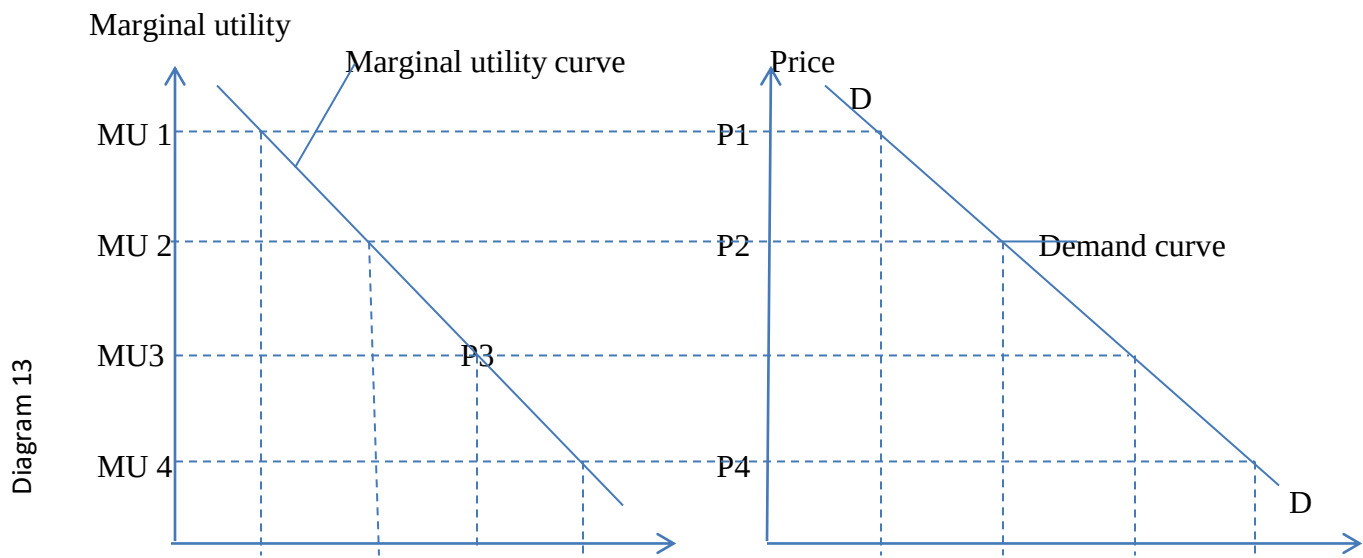
LIMITATIONS OF THE LAW OF DIMINISHING MARGINAL UTILITY

- It does not apply to commodities like diamond or hobby goods like paintings, stamp collection whose satisfaction increases as more is consumed e.g. the utility of additional paintings is greater than earlier pieces bought.
- It does not apply to indivisible durable commodities e.g. T.V sets, furniture etc whose consumption extends over a long period of time.
- It does not apply for habitual or addictive commodities e.g. cigarettes, alcohol, opium, etc, whose marginal utility may not diminish instead the more one takes, the more he/she will need it.
- It assumes homogeneity such that all units of a good should have the same quality and weight which is not the case.
- It assumes that units of a commodity should be consumed in succession. However if consumption for a commodity is spaced or at random, utility will increase and the law will not apply.
- It assumes there should be no change in habits, customs, fashion and income of consumers and when this happens the utility will increase instead of diminishing.

Relationship between the law of diminishing marginal utility and the demand curve

The relationship is that, as marginal utility falls, a consumer is willing to consume more units of a commodity at successively lower prices. Therefore, it is this law that explains the downward slope of a demand curve from left to right.

An illustration of the Marginal utility and the demand Curve:



0 1 2 3 4 qty0 1 2 3 4 Qty

SUB-TOPIC 3: THE THEORY OF SUPPLY:

SUPPLY:

This is the amount of goods and services the producers are willing to produce and put on market for sale at a given price and at a given period of time.

Quantity supplied is a desired flow i.e. it measures how much producers would like to sell and not how much they actually sell.

Individual supply; this is the quantity of a commodity that a firm/producer is willing to sell at various prices during a given time.

Market supply; this refers to quantities of a commodity that all producers are willing to offer for sale to a particular market at various prices during a given time.

TYPES OF SUPPLY:

1. COMPETITIVE SUPPLY: This is the supply of two or more commodities that use the same resource for their production such that an increase in the supply of one product leads to decline in the supply/production of the other. E.g.s include; eggs and meat from chicken, Milk and meat from cows, crop and animal production from a piece of land etc.

2. JOINT/COMPLEMENTARY SUPPLY: This refers to the supply of two or more commodities from the same process of production/same source such that an increase in supply of one commodity leads to an increase in supply of the other. E.g. Meat and skin from slaughtered animals, petrol, diesel and paraffin from crude oil through fractional distillation.

THE SUPPLY FUNCTION: This is a statement which shows a technical relationship between quantity supplied and the major determinants of quantity supplied. It is summarised as follows:

$$Q_s = f(P_1, P_2, \dots, P_n, F_n, G, T, \text{ etc})$$

Where: P_1 is the price of the commodity

$P_2, \dots, P_n$ is the price of the other commodity

$F_1, \dots, F_n$ are the factors of production

G----- Goal of the firm

T----- Level of technology

THE SUPPLY SCHEDULE: This is a table showing the quantity supplied of a commodity at various price levels over a period of time. It explains the law of supply which states that “the higher the price, the higher the quantity supplied of a commodity and the lower the price, the lower the quantity supplied of a commodity “*ceteris paribus*”/other factor being constant.

A table showing the supply schedule

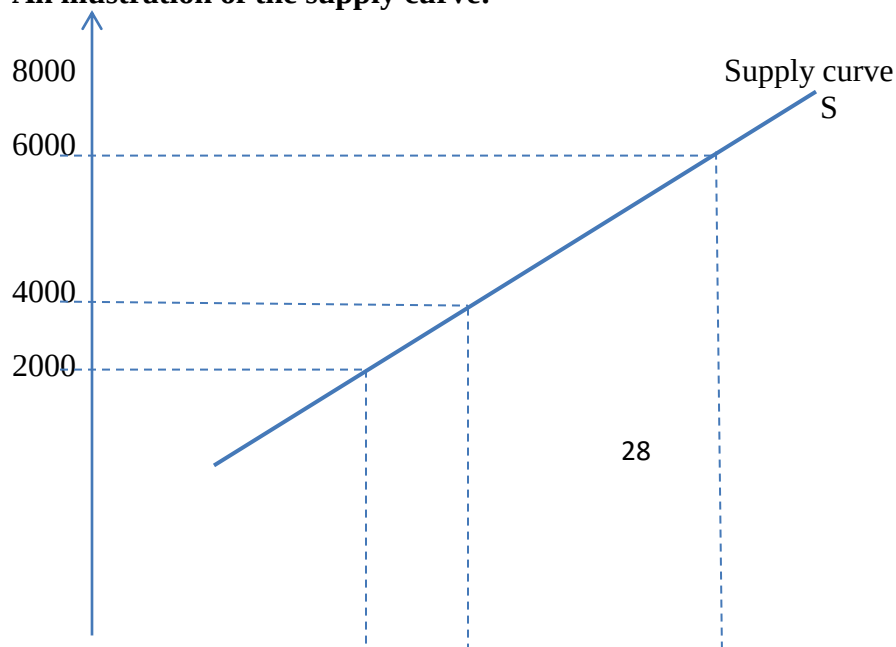
Price per unit (Shs.)	Quantity supplied per month in(kg)
2000	20,000
3000	40,000
4000	70,000
6000	100,000

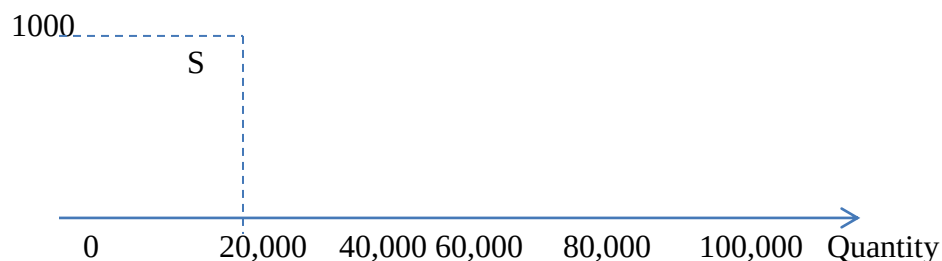
From the supply schedule, we can get the supply curve.

THE SUPPLY CURVE: This is a locus of points showing the quantity of a commodity supplied at different price levels in a given period of time i.e. it is a graphical representation of the supply schedule.

A supply curve slopes upwards from left to right indicating that more of a commodity is supplied at a higher price and less is supplied at a lower price.

An illustration of the supply curve:





THE SLOPE OF THE SUPPLY CURVE

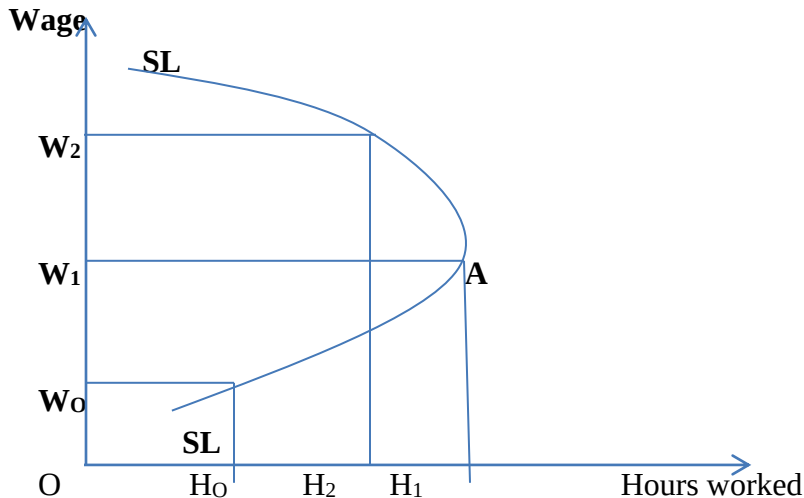
The supply curve is positively sloped i.e. it slopes from left to right showing the direct relationship between the price and quantity supplied of a commodity. The positive slope is explained by the following factors.

- i. **Entry of new firms in the industry.** When the price of a commodity increases, new firms are attracted to enter in the industry in order to enjoy the prospects of increasing profits, this leads to an increase in supply as price increases.
- ii. **Profit motive.** As producers aim at maximizing profits, they supply more at higher prices in order to increase the profitability of the business. A fall in price of the commodity reduces the quantity supplied because it is no longer profitable for them to sell more at lower prices.
- iii. **The struggle to maintain equilibrium in free market conditions.** As demand increases, price increases as well. Due to shortage, firms increase output in order to cover the shortage.
- iv. **Ease of diverting resources from the production of a commodity whose price has reduced to the production of a commodity whose price has increased.** For example, when the price of groundnuts increases, keeping the price of beans constant, producers easily divert resources i.e. land, labour, and capital, from production of beans to production of groundnuts. This leads to an increase in the supply of groundnuts as the price increases since producers expect higher profits.

ABNORMAL CURVES/REGRESSIVE/EXCEPTIONAL SUPPLY CURRVES: These are curves that do not obey/conform to the law of supply such that more of a commodity may be supplied at lower price or less of a commodity may be supplied at a higher price. There are two types of regressive supply curves namely: backward bending/slopping supply curve of labour and fixed supply.

1. An illustration of the backward bending/ slopping supply curve of labour:

Diagram 20



From the diagram above SL is the backward bending supply curve of labour.

When the wage rate increases from OW_0 to OW_1 labour supply increases from OH_0 to OH_1

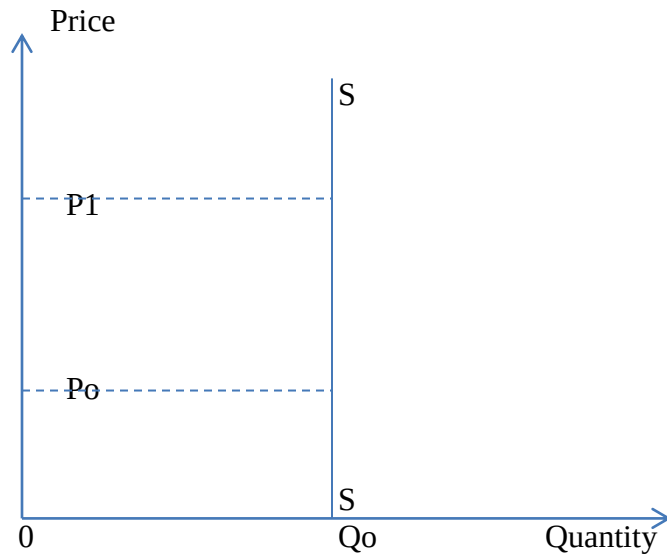
After point A workers start working for fewer hours because wage OW_1 was enough to meet their targets. At wage OW_2 labour supply reduces to OH_2

Causes of a backward slopping supply curve of labour:

- i. **Strong preference for leisure due to accumulated incomes.** Such a worker may prefer to work for less hours and spend most of the time enjoying leisure, even when the wage increases.
- ii. **The presence of target workers** i.e. some individuals get employed with certain aims or targets within their minds and once they have achieved them, they offer less hours of work even when the wage has increased.
- iii. **Increased levels of progressive taxation.** This because the harder the person works, the higher the amount of tax paid and therefore such a worker decides to work for less hours.
- iv. **Political instability in an area.** The fear of losing life due to insecurity makes people to work even when wage rises.
- v. **Reduction in real incomes.** The high general prices greatly reduce the purchasing power of the worker and see no reason for working harder even at a higher wage.
- vi. **Poor health of workers.** This weakens the worker and makes it hard for him to increase the number of hours worked even at a higher wage.
- vii. **Poor conditions of work.** These make the employees to hate the work and are not prepared to increase the number of hours even when the wage is increased.

2. Fixed supply curve. This is a situation where an increase in the price of a commodity does not have any effect on quantity supplied at all levels of price

An illustration of a fixed supply curve



In the figure above, despite the increase in price from OP_0 to OP_1 quantity supplied remains the same at OQ_0 .

FACTORS THAT AFFECT/ INFLUENCE/DETERMINE/ SUPPLY:

The favourable factors lead to high supply and the unfavourable natural factors lead to low supply.

1. **The price of a commodity;** high price of a commodity leads to high supply of a commodity, this is so because higher prices mean greater profits and so firms are attracted to supply high quantity of the commodity. On other hand low price of a commodity leads to low supply because of the low profits earned which discourages production hence low supply.
2. **Price of a jointly supplied good;** high price of a jointly supplied good leads to high supply of the commodity in question, this is so because the high price of the jointly supplied good makes it more profitable supply it, which leads to high supply of the commodity in question since they are both supplied from the same source/process of production. On the other low price of a jointly supplied good leads to low supply of the commodity in question, this is so because the low price of the jointly supplied good makes it less profitable to supply

it , which leads to low supply of the commodity in question because they are both supplied from the same source/ process of production.

3. **Price of a competitively supplied commodity/good.** High price of a competitively supplied good leads to low supply of the commodity in question; this is so because the high price of the competitively supplied good makes it profitable to supply it than the commodity in question. On the other hand low price of a competitively supplied good leads to high supply of the commodity in question, this is so because the low price of the competitively supplied good makes it less profitable to supply it and the suppliers opt for the commodity in question.
4. **The cost of production;** high cost of production leads to low supply of a commodity, this is so because the low profit margin enjoyed by the producers. On the other hand low cost of production leads to high supply of a commodity, this is so because of the high profit margin enjoyed by the producers
5. **The state of technology;** the use of advanced techniques /methods of production leads to high output/supply of a given commodity, this is so because of the high level of efficiency in production. On the other hand use of poor/ primitive/rudimentary techniques of production leads to low output/supply of a commodity, this is so because of the low level of efficiency in production.
6. **The goal/objective of the producer;** A producer whose aim is profit maximisation produces low output, this is so because he/she wants to restrict output and sell at high profit per unit sold. On the other hand a producer whose aim is sales maximisation produces high quantity of output; this is so because he wants to make profits by maximising sales.
7. **Government policy on production of the commodity;** Favourable government policy on production of a commodity in term of low taxes imposed on a commodity leads to high supply of a given commodity, this is so because it leads to low cost of production which implies high profits. On the other hand unfavourable government policy on production in terms of high taxes leads to low supply of a given commodities this is so because it leads to high cost of production which implies low profits.
8. **The gestation period of a commodity.** A long gestation period leads to low supply of a given commodity, this so because it take a long time to produce and supply such a

commodity. On other hand a short gestation period leads to high supply of a given commodity since it takes a short period to produce and supply such a commodity.

9. **Number of producers/ suppliers in the market;** A large number of producers leads to a high supply of a good, this is so because supply is from many producers, However a small number of producers leads to a low supply of a good since there are few producers to supply the commodity.
10. **Natural conditions/ factors.** Favourable natural factors lead to high supply of especially agricultural products; this is so because it favours the production of such products. However unfavourable natural conditions lead to low supply of especially of agricultural products because they discourage/limit their production.
11. **The political climate/ atmosphere;** Political stability leads to high supply of a given commodity, this is so because peace and stability enables people to engage in production of goods since they are not scared of losing their lives and property. However political instability leads to low supply of goods, this is so because it discourages production of goods, since people are scared of losing their lives and property.
12. **The market size/ demand for the commodity.** A large market size leads to high supply of a commodity; this is so because large market implies high profit margins thus motivating them to produce high output levels. On the other hand a small market size leads to low supply of a commodity because a small market size leads to low profit margins hence discouraging production.
13. **Level of entrepreneurial skills.** Presence of good entrepreneur skills leads to high supply of a commodity because they are many people to initiate businesses and sustain them. On the other hand low level of entrepreneurial skills leads to low supply of a commodity, this is so because there are few people to initiate businesses and sustain them.
14. **The land tenure system.** A favourable land tenure system promotes production leading to high supply of a given good; this is so because producers have easy access to land leading to high production levels. On the other hand poor land tenure system discourages production leading to low supply of a given good, this is so because of the difficulty in accessing land thus leading to low production levels.

15. **The state of infrastructure.** Good infrastructure for example good roads lead to high supply of a commodity, this because they lower the cost of transportation of goods to the market, on the other hand poor infrastructure for example bad roads lead to low supply of a good, this is because of the high cost of transportation of goods to the market which discourages production.
16. **Availability of factor inputs** e.g. raw materials, capital, skilled labour. Adequate supply of factor inputs such as capital leads to high supply of a good/commodity, this is so because the producers are able to purchase enough raw materials and produce goods and services. On the other hand limited supply factor inputs for example limited capital limits supply of a good because they cannot purchase enough raw materials which limits production.

Factors that lead to high supply of a commodity:

- High price of the commodity
- High price of a jointly supplied good
- Low price a competitively supplied good
- Low cost of production
- Advanced/High state of technology
- The goal of the producer being sales revenue maximisation
- Favourable government policy on production of a commodity\
- Short gestation period
- Large number of producers
- Favourable natural factors
- Political stability/Favourable political climate/atmosphere
- A large market size
- High level of entrepreneurial skills.
- Favourable land tenure system
- Good infrastructure
- Adequate supply of factor inputs

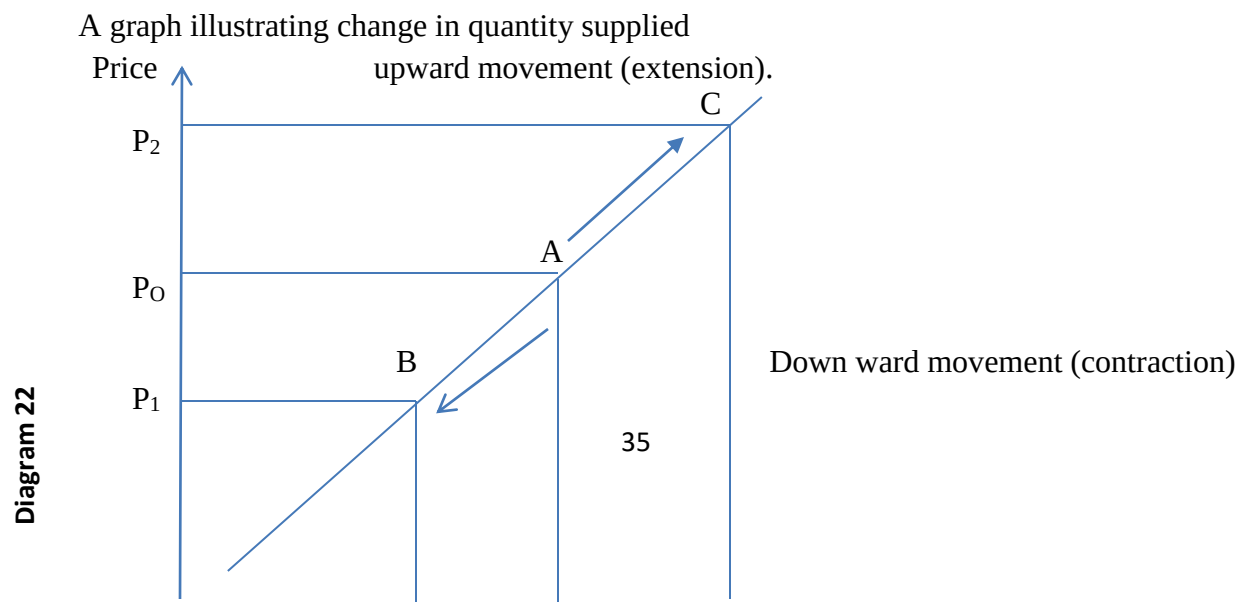
Factors that lead low supply of a commodity:

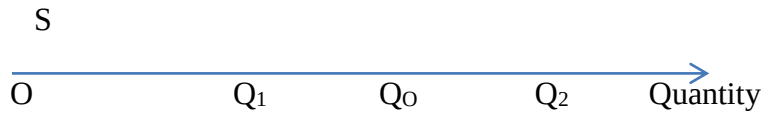
- Low price of the commodity
- Low price of a jointly supplied good\
- High price of a competitively supplied good
- High cost of production
- Poor state of technology
- The goal of the produce being profit maximisation
- Unfavourable government policy on production
- Long gestation
- Small number of producers
- Unfavourable natural factors/conditions
- Political instability/Unfavourable political climate/atmosphere
- A small market size.
- Limited/low levels of entrepreneurial skills
- Poor land tenure system
- Poor infrastructure
- Inadequate supply of factor inputs.

CHANGE IN QUANTITY SUPPLIED

This refers to an increase or decrease in the quantity supplied of a commodity due to the change in its price, when other factors that affect supply remain constant.

Change in quantity supplied is illustrated by movement along the same supply curve, either upward or downwards.





In the graph above, the downward movement along the same supply curve indicates a contraction in supply/ decrease in quantity supplied.i.e. (A to B). This is due to a fall in price from OP₀ to OP₁, indicating a decrease in quantity supplied from OQ₀ to OQ₁.

Note (i) Decrease in quantity supplied. This is a situation when less of a commodity is supplied due to a decrease in its price, when other factors that affect supply are held constant.

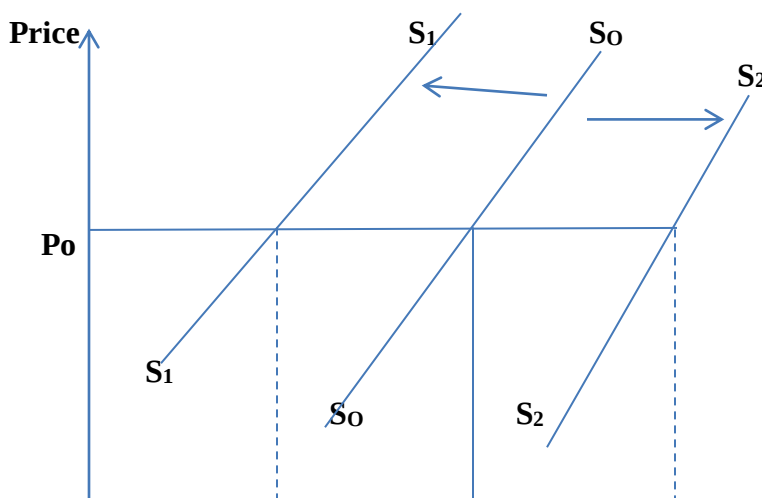
The upward movement along the same supply curve indicates an extension in supply/an increase in the quantity supplied. i.e.(A to C). This is due to an increase in price from OP₀ to OP₂ indicating an increase in quantity supplied from OQ₀ to OQ₂.

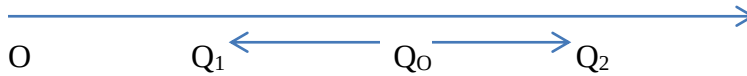
(ii)An increase in quantity supplied. This is a situation when more of a commodity is supplied due to an increase in its price, when other factors that affect supply remain constant.

CHANGE IN SUPPLY:

This refers to an increase or a decrease in supply of a commodity due to changes in the other factors that affect supply when the price of the commodity is constant.

A graph illustrating change in supply:





At each possible price (OP_0) in the graph above, quantity supplied can increase or decrease because of changes in other factors affecting the quantity supplied.

Increase in supply illustrated by the shift of the supply curve to the right i.e. S_0S_0 to S_1S_1 . Quantity supplied increases from OQ_0 to OQ_1 at a constant price OP_0 .

FACTORS THAT CAUSE A CHANGE IN SUPPLY OF A COMMODITY:

- **Change in the price of a jointly supplied good.** An increase in the price of a jointly supplied good leads to an increase in supply of the commodity in question, this is so because an increased price of the jointly supplied good makes it more profitable to supply it thus leading to an increase in supply of the commodity in question since they have the same procedure of production. On other hand a decrease in the price of a jointly supplied good leads to a decrease in supply of the commodity in question, this is so because the reduced price of the jointly supplied good makes it less profitable to supply it and thus a reduction in the supply of the commodity in question because they came from the same procedure of production.
- **Change in the price of a competitively supplied commodity/good.** An increase in the price of a competitively supplied good leads to a decrease in supply of the commodity in question; this is so because the producers shift the resources to the production of the competitively supplied commodity which is more profitable. On the other hand a decrease in the price of a competitively supplied good leads to an increase in supply of the commodity in question, this is so because the producers shift the resources to the production of the commodity in question which is more profitable.
- **Change in the cost of production;** an increase in the cost of production leads to a decrease in supply of a given commodity, this is so because of the reduced profit margin. On the other hand reduced cost of production leads to an increase in supply of a given commodity, this is so because of the increased profit margin enjoyed by the producers.
- **Change in the state of technology;** Improvement in the state of technology leads to increase supply, this is so because of the increased level of efficiency which encourages more

production. On the other hand decline in the state of technology leads to a decrease in supply; this is so because of the reduced level of efficiency which lowers the production.

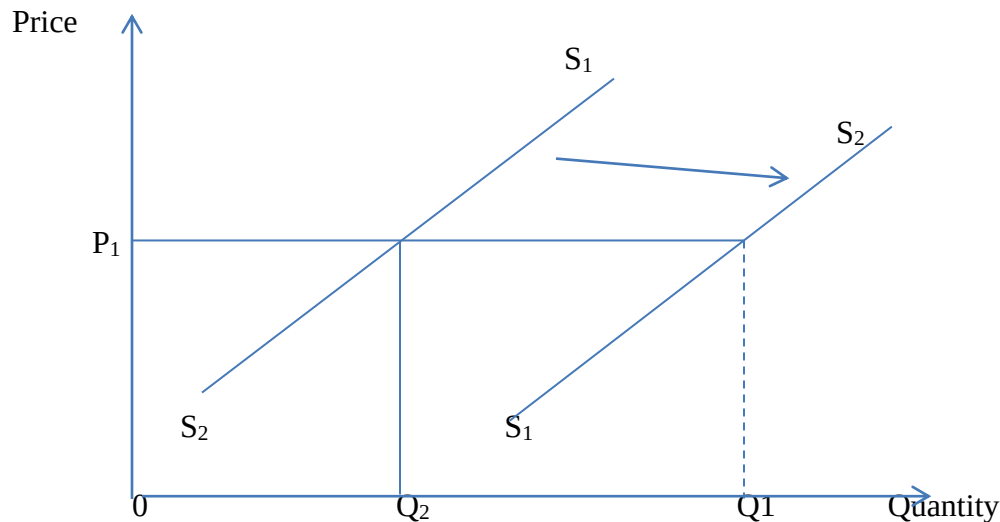
- **Change in the goal/objective of the producer;** A change in the goal of the producer from sales revenue maximisation to profit maximisation leads to decrease in supply of a given commodity, this is so because the producers want to supply less output to force an increase in price and make more profits. On the other hand the change in the goal of the producer from profit maximisation to sales revenue maximisation leads to an increase in supply of a commodity because he/she bring more on the market to increase his/her revenue.
- **Change in government policy on production of the commodity;** Favourable change in government policy on production of a commodity in term of reduced taxes imposed on a commodity leads to an increase in its supply; this is so because it leads to reduced costs of production which encourages production. On the other hand unfavourable change in government policy on production in terms of increased taxes leads to a reduction in supply of a given commodity, this is so because it leads to increased costs of production which discourages production of goods.
- **Change in the number of producers/ suppliers in the market;** an increase in the number of the producers leads to an increase in the supply of a given good; this is because it increases competition among the producers which forces them to increase output. On the other hand a decrease in the number of producers reduces supply because it lowers competition among producers leading to a fall in output levels.
- **Change in the natural conditions/ factors.** Favourable change in natural factor lead to an increase in supply of a given good especially an agricultural product; this is so because the crops and animals grower faster which increases agricultural output. On unfavourable change in natural conditions lead to a decrease in supply of especially of agricultural products because they destroy crops and animals which lead a reduction in output levels.
- **Change in the political climate/ atmosphere;** Favourable change in Political climate lead to increased supply of a given commodity, this is so because peace and stability enables people to engage increased production of a given good since they are not scared of losing their lives and property. However unfavourable change in political leads to reduced supply of

a given good, this is so because it discourages increased production of a given good, since people are scared of losing their lives and property.

- **Change in the market size/ demand for the commodity.** A reduction in the market size leads to a reduction in supply of a given commodity; this is so because it reduces the profit margins thus discouraging producers to increase output levels. On the other hand an increase in the market size leads to an increase in supply of a commodity because the increased market size leads to increased profit margin hence encouraging producers to increase output.
- **Change in the level of entrepreneurial skills.** Improved/increased level of entrepreneurship leads to increased supply of a given commodity, this is so because of increased ability to initiate businesses and sustain them. However a decline in the level of entrepreneurship leads to a decrease in the supply of a given commodity, this is so because of the reduced ability to initiate businesses and sustain them.
- **Change in the land tenure system.** An improvement in the land tenure system leads to increase in supply of a given good; this is so because producers have easier access to land leading to increased production levels. On the other unfavourable change in the land tenure system leads to a decrease in supply of a given good, this is so because it makes it more difficult to access land leading in production levels.
- **Change in the level of development of infrastructure.** Improvement in infrastructure for example construction of better roads increases supply because it lowers the transportation cost of goods and services which encourages production. On the other hand break down of infrastructure reduces supply of a commodity; this is so because it increases transportation costs of goods to the market which discourages production.
- **Changes in the supply of factor inputs for example change in the level of capital.** An increase in capital level increases supply because the producer is able to buy more raw materials and increase production. On the other a decrease in capital reduces supply because the producer buys fewer raw materials which lead to a decrease in production.

AN INCREASE IN THE SUPPLY OF A COMMODITY: This is a situation when more of a commodity is supplied due to favourable changes in other factors that affect supply when price is constant. It involves a shift of the supply curve to the right.

An illustration of an increase of a commodity.



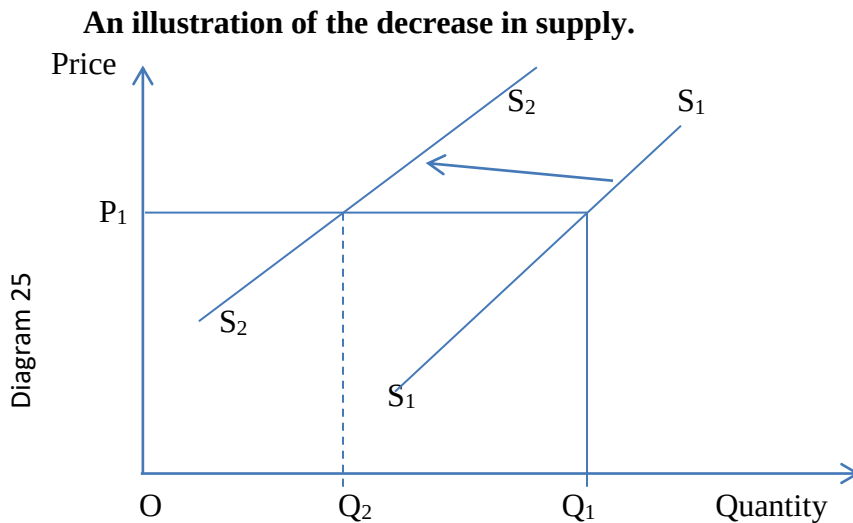
From the above illustration an increase in supply is indicated by the shift of the supply curve from S_1S_1 to S_2S_2 , leading to an increase in supply from OQ_1 to OQ_2 .

Factors that lead to an increase in supply of a commodity

- Natural factors becoming favourable
- An improvement in techniques of production.
- A fall/reduction in the cost of production.
- An increase in the number of suppliers in the industry.
- An increase in the market size./ An increase in the demand for the commodity.
- Reduction in the gestation period for a commodity.
- Change in the goal of the firm/producer from profit maximization to sales revenue maximisation.
- Favourable government policy on production of a commodity e.g. reduced taxes/ Increased subsidisation.
- An improvement in the political climate.
- An improvement in the entrepreneurial skills
- An improvement in the land tenure system.
- An improvement in the state of infrastructure

- An increase in the price of the jointly supplied good
- A decrease in the price of a competitively supplied good
- Breakdown in infrastructure
- Reduction in supply of factor inputs
- Reduction in the gestation period

DECREASE IN SUPPLY: This refers to a situation when less of a commodity is supplied due to unfavourable changes in other factors that affect supply when price is constant. It is indicated by a shift of the supply curve to the left.



From the illustration above the decrease in supply is indicated by a shift of the supply curve S₁S₁ to S₂S₂ leading to a decline in supply from OQ₁ to OQ₂.

Factors that lead to a decrease in supply of a commodity.

- An increase in the cost of production
- Natural factors/conditions becoming unfavourable
- A decline in the level of technology./methods of production
- A reduction in the number of suppliers/producers in an industry/Exit of some firms from the industry.
- Political atmosphere/ climate becoming unfavourable
- A decrease in the market size/ a decrease in the demand for a commodity
- Change in the goal of a firm/producer from sales revenue maximization to profit maximisation.
- Government policy on the production of the commodity becoming unfavourable e.g. increased taxes/reduced subsidies.
- A fall in the entrepreneurial skills/ability
- Breakdown of infrastructures
- Decrease in the price a jointly supplied product
- Increase in the price of a competitively supplied commodity.

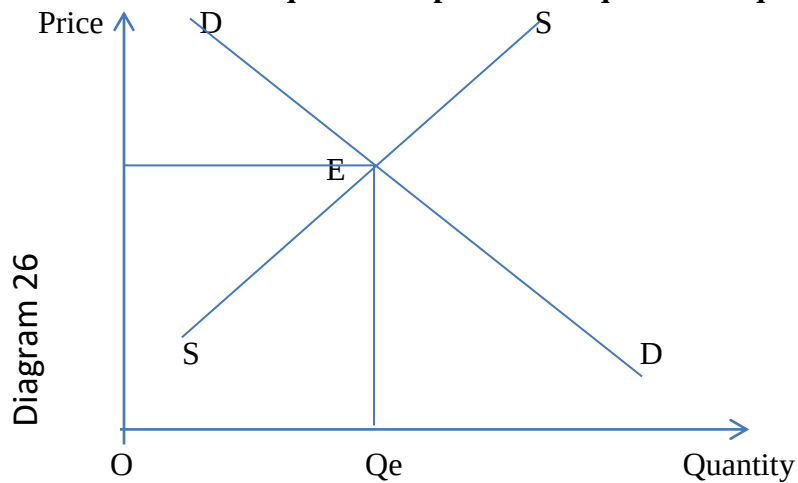
SUB-TOPIC 4: MARKET EQUILIBRIUM:

EQUILIBRIUM PRICE: This is price established in the market when quantity demanded is to equal quantity supplied of a commodity. Equilibrium price also changes from time to time. It is also known as **clearing price**. i.e. it is set or fixed at a point of intersection of demand and supply curves in a free enterprise economy.

EQUILIBRIUM QUANTITY: This is the quantity exchanged when the market is in balance.i.e. quantity demanded and quantity supplied are equal, therefore there is no shortage or surplus in the market which means that neither buyers or sellers are inclined to change the price or the quantity which is an essential condition for equilibrium.

The only quantity that accomplishes this task is at the intersection of the demand curve and supply curve.

An illustration of Equilibrium price and Equilibrium quantity:



In the illustration above, point E is the equilibrium point and OPe represents the equilibrium price while OQe represents the equilibrium quantity.

Deriving the market equilibrium using the demand and supply functions:

Market equilibrium is derived by using the demand function and the supply function where quantity demanded (Q_d) is equal to quantity supplied (Q_s)

When solving for equilibrium price and quantity you need to have a demand function and a supply function.

E.g. if your monthly demand function is $Q_d = 10000 - 80P$ and your monthly quantity supply function is $Q_s = 20P$, Then set $Q_d = Q_s$ and solve.

Solution:

$Q_d = Q_s$
i.e. $10000 - 80P = 20P$
 $10000 - 80P + 80P = 20P + 80P$
 $10000 = 100P$
Therefore;
 $P = 10000/100$
 $= 100$.
Equilibrium price = 100

NB: Equilibrium price has no units; this is because it is hypothetical/theoretical.

To find equilibrium quantity we substitute equilibrium price (100) into either the demand function or supply function as follows;

(a) Using the demand function

$$Q_d = 10000 - 80P$$

$$= (10000) - 80 \times 100$$

$$= 10000 - 8000$$

$$= 2000.$$

(b) Using the supply function:

$$Q_s = 20P$$

$$= 20 \times 100$$

$$= 2000$$

Therefore $Q_d = Q_s$ (as shown by the above substitution above):

Exercise:

Given that (i) $Q_s = -12 + 12p$ and $Q_d = 36 - 4P$

(ii) $Q_d = 24 - 4P$ and $Q_s = -8 + 8P$

(iii) $Q_d = 24 - 2P$ and $Q_s = -12 + 4P$

Calculate the ;(a) Equilibrium price

(b) Equilibrium quantity.

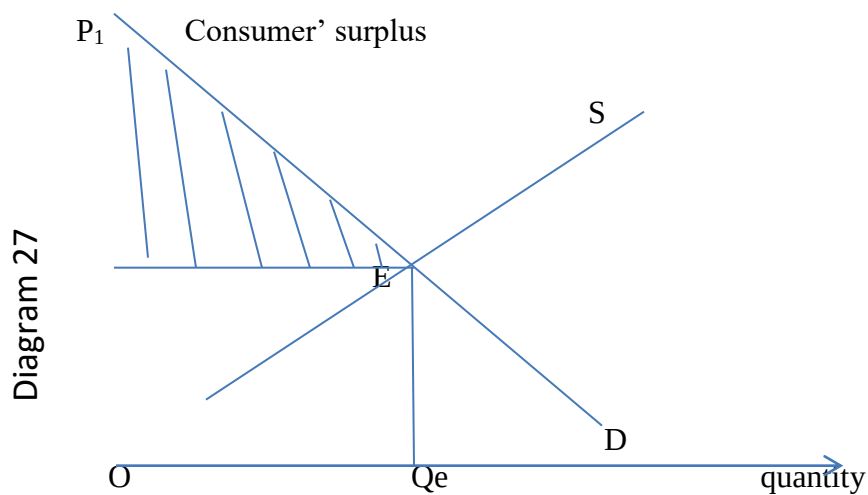
THE CONCEPT OF CONSUMER'S SURPLUS: This refers to the difference between what a consumer is willing to pay and what he/she actually pays for the commodity.

OR: It is the extra utility/ additional satisfaction a consumer enjoys without paying for it

It is given by the following formula:

NOTE: Consumer's surplus is limited by: income inequality among consumers, changing marginal utility of money, differences in tastes and preferences and presence of goods of ostentation

Illustration of the consumer surplus



The is shown as the area under the demand curve and the prevailing market price

In the illustration above if the consumer purchases a commodity at OP_e , we note the following;

- Total utility is equal to OP_1EQ_e
- Expenditure by the consumer is OP_eEQ_e

The consumer surplus is illustrated by the shaded area below the demand curve above the market price at which the consumer buys the commodity i.e. OP_1EP_e

Example:

Using the consumer's demand schedule below, calculate the consumer surplus when the market price is shs.150.

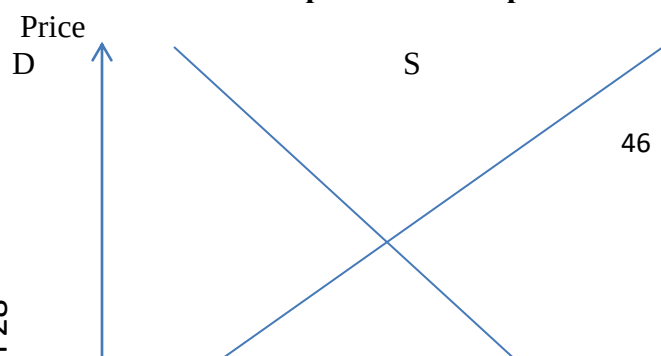
Price consumer is willing to pay	300	250	200	150	100	50
Units purchased	1	2	3	4	5	6

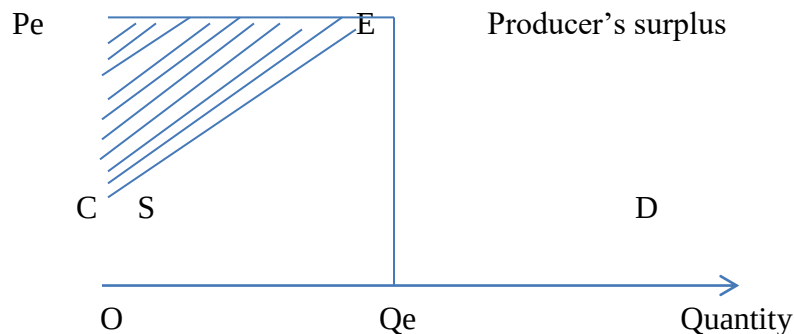
Consumer surplus = Planned expenditure - Actual expenditure
 But actual expenditure = Market price $\times$ Number of units purchased.
 = $(300+250+200+150) - (150 \times 4)$
 = Shs. 900 - 600
 = Shs. 300

PRODUCER'S SURPLUS: This refers to the difference between what the producer actually charges for the commodity and what he was willing to charge.

Producer's surplus = **Actual revenue** – **Expected revenue.**

An illustration of the producer's surplus.





In the illustration above the producer's surplus is shown by the shaded area above the supply curve and below the equilibrium price.i.e. OPeEC.

Example:

Price per unit	300	350	400	450	500	550	600	650
Quantity supplied	1	2	3	4	5	6	7	8

Given the market price is shs.550. Calculate the producer surplus if 6 units are sold.

$$\begin{aligned}
 \text{Producer surplus} &= \text{Actual revenue} - \text{Planned revenue} \\
 &= (550 \times 6) - (300 + 350 + 400 + 450 + 500 + 550) \\
 &= 3300 - 2550 \\
 &= \underline{\text{Shs. 750}}
 \end{aligned}$$

SUB-TOPIC 5: THE CONCEPT OF ELASTICITY:

Elasticity refers to the measure of the degree of responsiveness of the dependent variables to the independent variables

Dependent variables may be the quantity demanded or quantity supplied while independent variables are the factors which influence the above dependent variables e.g. Price of the commodity
Price of other commodities, consumer's income

Elasticity can be taken to mean reaction or response of producers or consumers.

Elasticity is of two types:

- a. Elasticity of demand
- b. Elasticity of supply

ELASTICITY OF DEMAND: This refers to a measure of the degree of responsiveness of quantity demanded of a commodity to changes in the factors that affect demand. E.g. price of the commodity, Consumer's level of income and price of related commodities.

Elasticity is considered for only three major determinants of demand namely:

- Price of the commodity in question
- Price of related commodities
- The level of income of the consumer

Therefore there are **three** different concepts of elasticity of demand namely

- Price elasticity of demand
- Cross elasticity of demands
- Income elasticity of demand

PRICE ELASTICITY OF DEMAND: This refers to a measure of the degree of responsiveness of quantity demanded of a commodity to changes in its price. i.e. it shows how much quantity demanded of a commodity responds to a change the price of that commodity.

P.E.D= (-)Proportionate change in quantity demanded of a commodity
Proportionate change in the price of the commodity

OR:

P.E.D =(-)Percentage change in quantity demanded of a commodity
Percentage change in the price of a commodity

OR: $PED = \frac{-\Delta Q}{\Delta P} \times \frac{P}{Q}$

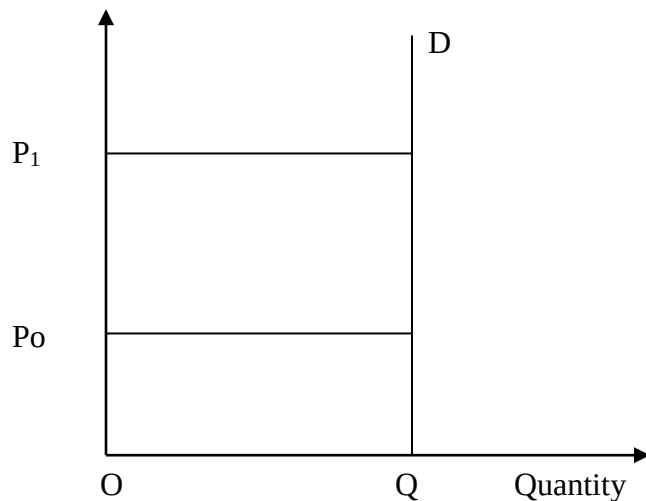
NB: A negative sign is multiplied into the formula to make the answer positive, since price elasticity of demand never is negative.

INTERPRETATION OF PRICE ELASTICITY OF DEMAND:

1.PERFECTLY /COMPLETELY INELASTIC DEMAND:.. Perfectly inelastic demand is one where price changes have no effect at all on quantity demanded of a commodity. This means that

quantity demanded remains constant in spite the price changes. The coefficient is equal to zero ($PED = 0$).

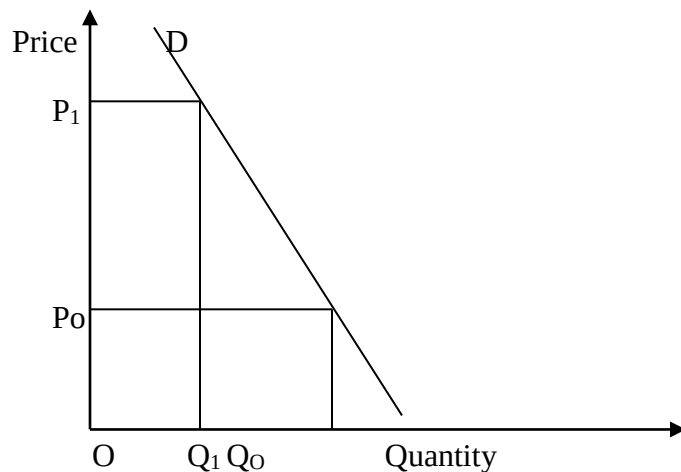
A graph illustrating a perfectly inelastic demand curve:



In the graph above, changes in price from OP_0 to OP_1 leaves the quantity demanded of a commodity constant at OQ_0 e.g. demand of cigarettes.

2. **FAIRLY INELASTIC DEMAND.** This is one where a very big percentage change in price leads to a small percentage change in quantity demanded of a commodity. The coefficient is greater than zero but less than one ($PED > 0 < 1$).

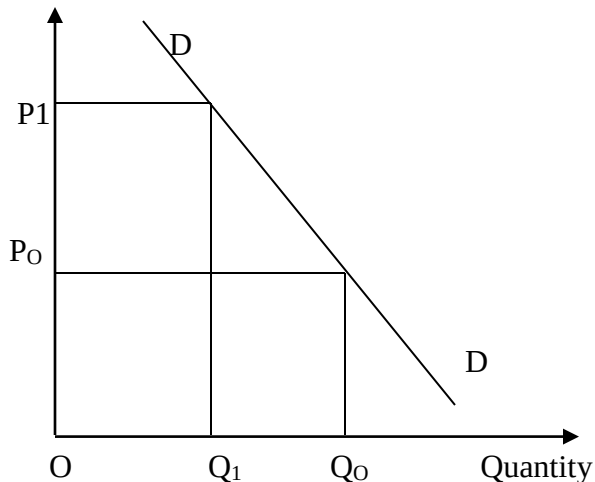
A graph illustrating a fairly inelastic demand curve:



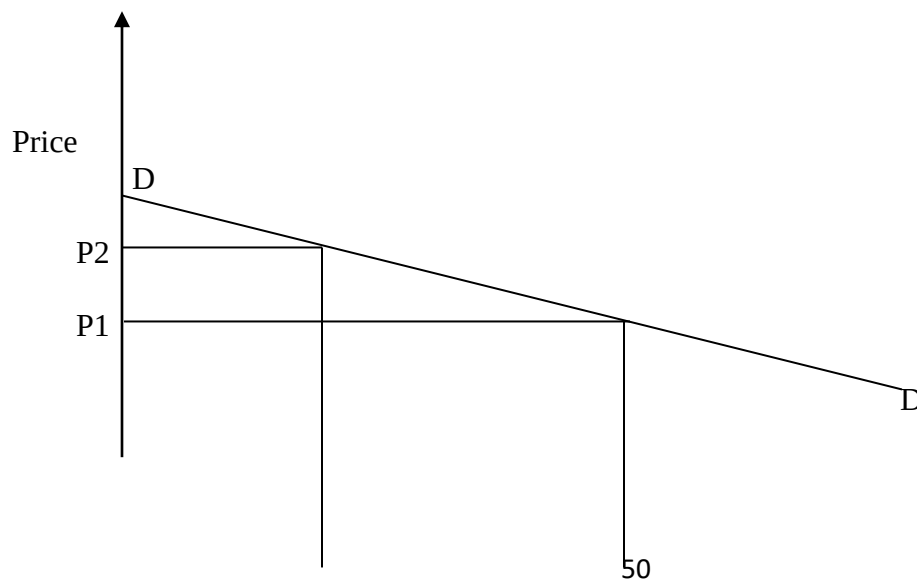
In the graph above a very big percentage in price e.g. OP_0 to OP_1 , leads to a very small percentage change in quantity demanded from OQ_0 to OQ_1 .

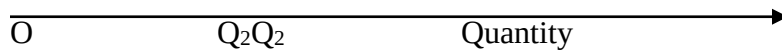
Unitary elasticity of demand. Unitary elasticity of demand is one where the percentage change in price is exactly equal to percentage change in quantity demanded of a commodity. The coefficient is equal to one ($PED = 1$).

An illustration of a unitary elasticity of demand:



Fairly elastic demand/ Elastic demand. This is one where a very small percentage change in price leads to a big percentage change in quantity demanded of a commodity. The coefficient is greater than one but less than infinity ($PED > 1 < \infty$).

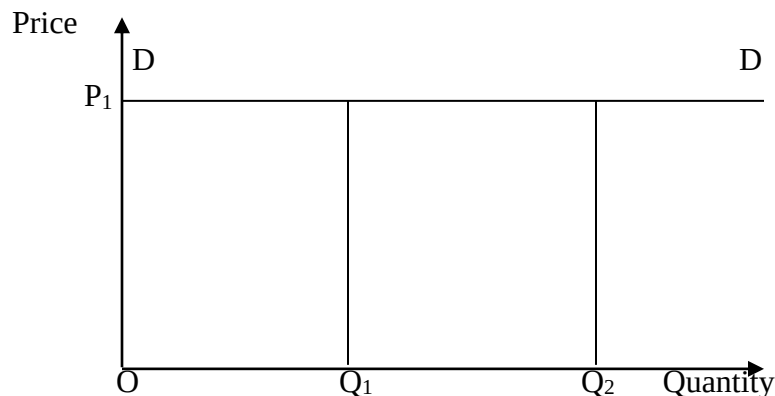




In the graph above a small percentage in price from OP1 to OP2 leads to a very big percentage change in quantity demanded of a commodity from OQ1 to OQ2.

Perfectly elastic demand. This is one where at a prevailing price consumers are willing to purchase the commodity infinitely.(as much as they want) and none at all at even a slightly higher price. The coefficient is equal to infinity ($PED = \infty$).

An illustration of a perfectly elastic demand curve:



In the graph above consumers are willing to buy as much as they want at the prevailing price i.e. OP1 and none at all even at all even a t a slightly higher price.

FACTORRS DETERMINING/AFFECTING/INFLUENCING PRICE ELASTICITY OF DEMAND

1. **Availability of substitutes;** a commodity with close substitutes has price elastic demand because an increase in its price leads to a big reduction in quantity demanded, because consumers have cheaper alternatives to turn to. On the other hand a commodity with no close substitutes has price inelastic demand because as its price increases, consumers continue buying almost the same quantity because they have no cheaper alternatives to turn to.
2. **Degree/Level of necessity of the commodity;** demand is price inelastic for essential goods (necessities) because with increase in price of a commodity consumers continue buying almost the same amount since they cannot do without them. On the other hand demand for luxuries/non essential goods is price elastic because with an increase in price of such a good consumers reduce its consumption since they can do without it.
3. **Level of durability/Perishability of the commodity;** the demand for a durable commodity is price inelastic because as its price reduces, the consumer does not buy, this

is because it can be used for a long time without replacement. On the other hand, the demand for a perishable commodity is price elastic; this is because with a fall in its price more of it is bought because it cannot be kept for a long time hence needs constant replacement.

4. **The proportion of income spent on the commodity;** the demand for a commodity that takes a small proportion of the consumer's income is price inelastic, this is so because as its price increases, the consumer continues buying it since he does not feel the pinch. On the other hand the demand for a commodity that takes a big proportion of the consumer's is price elastic; this is so because as its price increases, the consumer reduces consumption since he/she feels the pinch.
5. **Level of addiction in the use of the commodity;** demand for a commodity consumed out a habit is price inelastic because with an increase in its price, the consumer continues to buy almost the same amount since it is not easy to break/abandon the habit. On the other hand the demand for a commodity consumed not out of a habit is price elastic, because as its price increases, the consumer reduces its consumption since he/she easily abandons the commodity since there is no special attachment to it.
6. **Level of income of the consumer;** the demand for a commodity is inelastic for a high income earner because with the increase in its price, the consumer continues buying the same amount of the commodity, since he/she can afford to buy it at whatever price. On the other hand, the demand is elastic for a low income earner because as the price increases the consumer reduces the amount bought since the purchasing power reduces.
7. **The number of uses of the commodity;** the demand for commodity with several uses is price elastic because with an increase in its price, the consumer reduces on some of the uses and remains with only the very essential ones. On the other hand the demand for commodity with a single use is price inelastic, because as the price increases the consumer maintains the quantity bought because he/she has no chance of reducing the quantity bought.

8. Time period; the demand for a commodity is price inelastic in the short run, because with increase in its price, the consumer continues buying, since the time is too short to find a cheaper alternative. On the other hand the demand for a commodity is price elastic in the long run, because with increase in price of a commodity the consumer reduces the amount consumed of it since the time is long enough to change the habit or find a cheaper substitute/alternative.

9. Season; the demand for a commodity during a favourable season is price inelastic, this is so because with an increase in its price, the consumer continues buying such a commodity due to the apparent need for it. On the other hand the demand for a commodity during unfavourable season is price elastic; this is because with a reduction in

the price of the commodity, the consumer reduces the amount consumed of it due to the reduced apparent need for the commodity.

10. Possibility of postponement of demand for the commodity; the demand for goods, the use of which can be postponed is price elastic, this is so because as their prices increase consumers reduce their consumption since their use is not very urgent. On the other hand demand for a good, the use of which cannot be postponed is price inelastic, this is because as their prices increase consumers continue buying them because their use is very urgent.

11. Level of advertising; the demand for commodity which is highly advertised is price inelastic because with an increase in its price the consumer continues buying since he/she is convinced that the product is better than all the others. On the other hand the demand for a commodity which is not intensively advertised is price elastic, because as the price increases the consumer drops the demand since he is not convinced to buy.

12. Availability of complements; high degree of Complementarity makes the demand price inelastic because as the price increases the demand for the commodity is maintained since the consumption for the commodity is combined with many others. On the other hand low degree of Complementarity makes the demand price elastic because as the price increases quantity demanded reduces because its consumption is not combined with any other and the consumer easily abandons it.

13. Degree of convenience in obtaining a commodity; the demand for a commodity which is not convenient to the consumer to obtain is price elastic because as the price increases the consumer quickly abandons it since it is burdensome for him/her to obtain. On the other hand the demand for a commodity that is convenient to obtain is price inelastic because even when the price increases, the consumer maintains the quantity bought since the commodity is easily accessible.

Factors Responsible For Inelastic Demand for a Commodity:

- Absence of close substitutes/ Low degree of substitutability
- High degree of Complementarity commodities/ presence of complements.
- High degree of necessity.
- High degree of addiction/ habit forming
- High degree of durability of the commodity
- Small proportion of income spent on a commodity.
- High level of consumer's income.
- Few uses of a commodity.
- High level/degree of convenience of getting a commodity.
- Short-run period.
- Expectation of future price increase.
- Favourable season
- High level of advertising

- High degree of urgency in the use of a commodity/ Inability to postpone the consumption of the commodity.

Factors Responsible for Elastic Demand for a Commodity:

- Presence of close substitutes/ High degree of substitutability
- Low degree of Complementarity
- Low degree of necessity/ demand for luxuries..
- Low degree of addiction in the use of a commodity.
- High degree of Perishability of the commodity
- Big proportion of income spent on a commodity
- Low level of consumer's income
- Many/numerous uses of a commodity.
- Low level/degree of convenience of getting a commodity
- Long-run period.
- Expectation of future price fall
- Unfavourable season
- Low level of advertising.
- Low degree of urgency in getting a commodity.

IMPORTANCE/USES OF PRICE ELASTICITY OF DEMAND TO THE GOVERNMENT:

- **It is used in taxation policy;** high taxes are imposed on commodities with inelastic demand while low taxes are imposed on commodities with elastic demand.
- **It is used in subsidisation policy;** Government grants subsidies to commodities with elastic demand and denies subsidies to commodities with inelastic demand
- **It is used in nationalisation policy;** government nationalises firms/enterprises that produce essential goods whose demand is inelastic. .
- **It is used in the devaluation of a country's currency;** devaluation is successful where demand for both imports and exports is price elastic, on the other hand it is less effective where demand is price inelastic.
- **It is used in wage determination;** government pays high wages to employees whose demand is price inelastic and pays low wages to employees whose demand is price elastic.

IMPORTANCE/USES OF PRICE ELASTICITY OF DEMAND TO THE PRODUCER:

- **It is used as a basis of price discrimination;** the producer/a monopolist charges a high price in a market whose demand is price inelastic and charges a lower price in the market whose demand is price elastic.
- **It is used in wage determination;** employers pay high wages to employees whose demand is inelastic and pay low wages to employees whose demand is price elastic.

- **It is used in pricing of commodities;** producers charge a high price for commodities whose demand is price inelastic and charges low prices to commodities whose demand is price elastic.
- **It is used in determining incidence of a tax;** the incidence falls more on the consumer where the demand for the commodity is inelastic and it falls more on the producer where the demand is elastic.

INCOME ELASTICITY OF DEMAND:

This refers to a measure of the degree of responsiveness of quantity demanded of a commodity to changes in the income of a consumer.

Mathematically it is given by the following formula:

Y.E.D = proportionate change in quantity dd of a commodity

Proportionate change in consumer's income

i.e. $\frac{(\text{New demand} - \text{old demand})}{\text{old demand}}$

OR:
$$YED = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

OR: Y.E.D= percentage change in quantity demanded of a commodity

Percentage change in consumer's income

OR:
$$YED = \frac{\% \Delta Q}{\% \Delta Y}$$

Example

Assuming that a person's salary increased from Shs15,000 to Shs20,000 and quantity demanded of a commodity decreased from 10 kg to 6 kg. Calculate income elasticity of demand.

$$YED = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

$$YED = \frac{6-10}{20000-15000} \times \frac{15000}{10}$$

$$YED = -1.2$$

Interpretation of coefficients of income elasticity of demand

- Negative coefficient implies it is an inferior good.
- Positive coefficient implies it is a normal good.

- Negative coefficient implies it is a zero.
- **Income elastic.** This is when the income elasticity of demand is greater than one, meaning that quantity demanded changes proportionally more than change in income. A slight change in the consumer's income leads to a very change in quantity demanded
- **Income inelastic.** This is when income elasticity of demand is less than one but greater than zero. This means that a large percentage change in income leads to a proportionately less percentage change in quantity demanded.

Importance of income elasticity of demand

- i. It is used in taxation policies.
- ii. It is used to determine the nature of goods
- iii. It is used to forecast future demand for commodities as level of income changes.

CROSS ELASTICITY OF DEMAND: This is the measure of the degree of responsiveness of quantity demanded of one commodity to changes in price of other commodities/ related commodities

$$\text{Cross elasticity of demand} = \frac{\text{Percentage change in quantity demanded of X}}{\text{Percentage change in price of commodity Y}}$$

$$\text{CED} = \frac{\% \Delta Q_x}{\% \Delta P_y}$$

$$\text{OR: C.E.D} = \frac{\text{Proportionate change in quantity demanded of commodity X}}{\text{Proportionate change in price of commodity Y}}$$

$$\text{CED} = \frac{\Delta Q_x}{\Delta P_y} \times \frac{P_y}{Q_x}$$

Example:

Given that the price of commodity Y increased from Shs. 100 to Shs. 150 and quantity demanded of a related commodity X increased from 50 kg to 90 kg. Calculate the cross elasticity of demand.

$$\text{CED} = \frac{\Delta Q_x}{\Delta P_y} \times \frac{P_y}{Q_x}$$

$$\text{CED} = \frac{90-50}{150-100} \times \frac{100}{50}$$

$$\text{CED} = 1.6$$

Interpretation of coefficients of cross elasticity of demand

- i. Positive coefficient implies the two commodities are substitutes
- ii. Negative coefficient implies the two commodities are complements
- iii. Zero coefficient implies that the two commodities are independent/ not related at all.
- iv.

Importance/uses of cross elasticity of demand:

- i. Used to classify commodities.
- ii. Used in the pricing policy.

ELASTICITY OF SUPPLY: This refers the measure of the degree of responsiveness of quantity supplied of a commodity to changes in factors that influence supply such as price of the commodity.

PRICE ELASTICITY OF SUPPLY: This refers to the measure of the degree of responsiveness of quantity supplied of a commodity to changes in its own price.

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied of a commodity}}{\text{Percentage change in price of a commodity}}$$

$$\text{PES} = \frac{\% \Delta Q_s}{\% \Delta P}$$

$$\text{OR: PES} = \frac{\text{Proportionate change in quantity supplied of a commodity}}{\text{Proportionate change in price of commodity}}$$

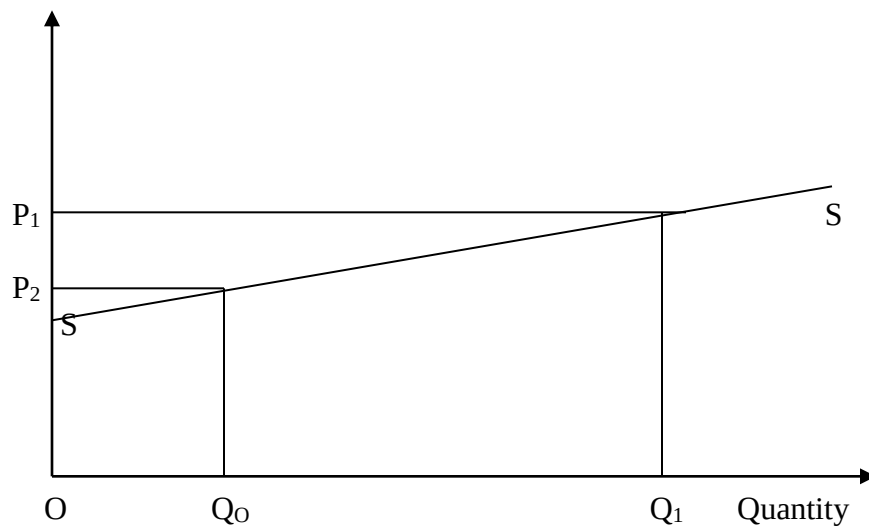
$$\text{PES} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Price elasticity of supply has a positive coefficient.

Categories of Price Elasticity Of supply

Fairly elastic supply. This is one where a very small percentage change in price leads to a big percentage change in quantity supplied of a commodity. The coefficient is greater than one but less than infinity ($\text{PES} > 1 < \infty$).

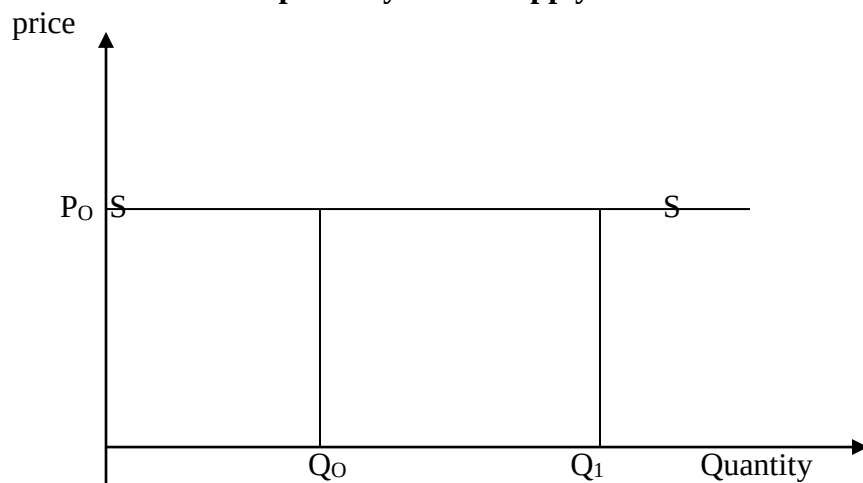
An illustration of a fairly elastic supply curve.



From the diagram above a small percentage change in price(OP_0 to OP_1) leads to a very big percentage change in quantity supplied of the commodity.(OQ_0 to OQ_1)

2. Perfectly elastic supply; This is one where at a prevailing price or above that producers are willing to supply all they can and none at all below that price. The coefficient is equal to infinity ($PES = \infty$).

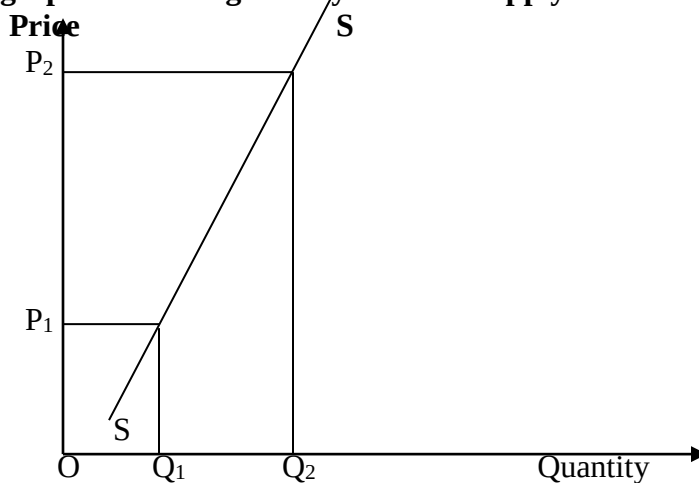
An illustration of a perfectly elastic supply curve.



From the graph above producers are willing to supply all they can at price OP or above price and none at all even at a slightly lower price.

3. Fairly inelastic supply. Inelastic supply is one where a very big percentage change in price leads to a small percentage change in quantity supplied of a commodity. The coefficient is greater than zero but less than one ($PES > 0 < 1$).

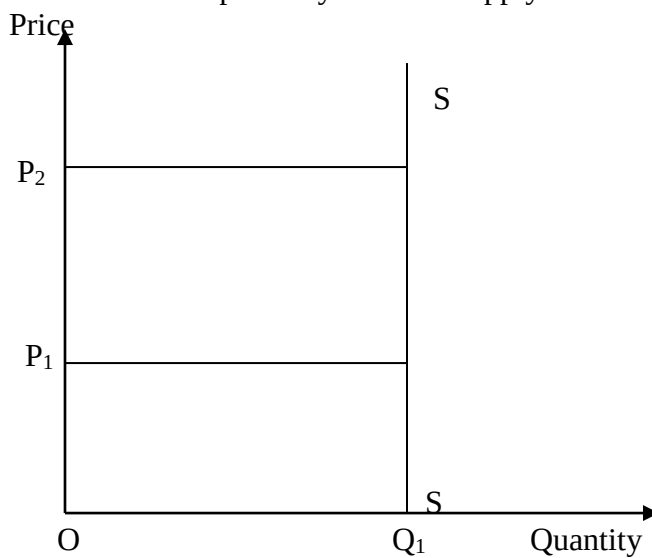
A graph illustrating a fairly inelastic supply curve:



From the illustration above a very big percentage change in price (OP_1 to OP_2) leads to a very small percentage change in quantity supplied (OQ_1 to OQ_2).

4. Perfectly inelastic supply. Perfectly inelastic supply is one where price changes have no effect at all on quantity supplied of a commodity. This means that quantity supplied remains constant in spite the price changes. The coefficient is equal to zero ($PES = 0$).

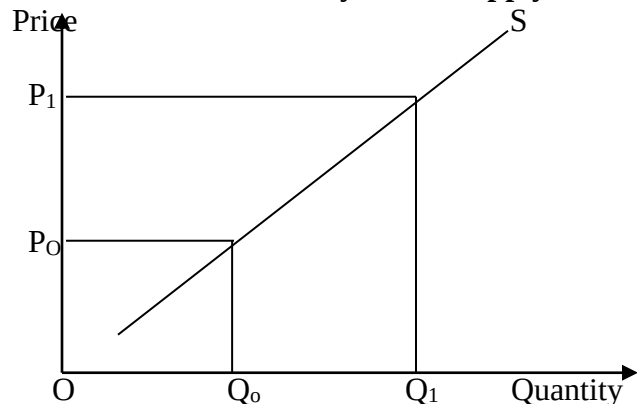
An illustration of a perfectly inelastic supply curve:



From the illustration above changes in price from OP1 to OP2 leaves quantity supplied unaffected at OQ1.

5. Unitary elasticity of supply. Unitary elasticity of supply is one where the percentage change in price is exactly equal to percentage change in quantity supplied of a commodity. The coefficient is equal to one ($PES = 1$).

An illustration of unitary elastic supply curve:



From the illustration above the percentage change in price from OPo to OP1 is equal to the percentage change in quantity supplied from OQo to OQ1

Factors that affect/influence/ determine Price Elasticity of Supply:

- **The cost of production:** The supply of a commodity is price inelastic when the costs of production are high; this is so because with increase the price of the commodity, producer is reluctant to increase supply, due to the low profit margin. On the other hand supply of a commodity is price elastic when the costs of production are low, this is so because with increase in the price of the commodity, producers quickly supply due to high profit margin.
- **Gestation period.** The supply of a commodity with a short gestation period is price elastic, this is so because with increase in the price of such a commodity, producers in position to increase supply within a short period of time. On the other hand the supply of a commodity with a long gestation period is price inelastic in the short run, this is so because with increase in price of such a commodity, producers are not able to increase supply because the production process take some time.
- **The state of technology.** The supply of a commodity produced using advanced technology is price elastic; this is so because with increase in the price of such a commodity, the producers are to increase the supply of such a commodity due to the use of such modern technology. On the other hand the supply of a commodity produced using poor technology is price inelastic, this is so because with increase in the price of such a commodity, the producers are not able to increase supply due to the inefficiency of such technology

- **Perishability/Durability of a commodity:** The supply of perishable commodities is price inelastic, this is so because with increase in the price of such a commodity, the producers cannot increase supply once existing stock have been exhausted since such goods cannot be stored for a long time. On the other hand the supply of durable commodities is price elastic, this is so because with increase in the price of such goods, the producers are able to supply by getting from the stores since the goods can be stored from the stores.
- **Time period.** The supply of a commodity is price inelastic in the short run. This is so because with increase in the price of the commodity, the producers cannot easily increase the supply of the commodity, since the time is too short to increase the supply. On the other hand the supply of a commodity is price elastic in the long run period, this is so because with increase in the price of the commodity, the producers can easily increase supply since the time is long enough to allow production and supply of more goods.
- **Natural factors especially in agricultural sector.** The supply of a commodity is price elastic during favourable natural factors such as adequate rainfall, fertile soils; this is so because with an increase in price of a commodity, the producers increase production and supply of a commodity since the conditions are favourable for the activity. On the other hand the supply of a commodity during unfavourable natural factors such as prolonged drought, infertile soils is price inelastic, this is so because with increase in the price of the commodity the producers are not able to increase production and supply of the commodity since the conditions are unfavourable for the activity.
- **Degree of freedom of entry of firms in production/Ease of entry of new firms in the industry.** The supply of a commodity is price elastic when there is freedom of entry of firms in an industry; this is so because with increase in the price of such a commodity it induces other firms to join production since there are no restrictions on entry. On the other hand the supply of a commodity is price inelastic when there is restricted entry into the industry; this is so because with increase in the price of a commodity is not easily increased since there are few producers due to restricted entry into the industry.
- **Political climate.** The supply of a commodity is price elastic when there is political stability this is so because with increase in the price of the commodity, the producers easily increase production and supply of a commodity, since such situations allow producers to produce more output as they live a settled life. On the other hand supply of a commodity is price inelastic; this is so because with an increase in the price of the commodity, the producers are not in position to increase supplies of the commodity since such situation do not allow production to easily take place as people do not live a settled life.
- **Future price expectations.** The supply of a commodity is price inelastic when there is an expectation of a further future price increase this is so because with an increase in the price of the commodity, the producers are reluctant to supply more of the commodity since they want to supply more in future at a much higher price. On the other hand the supply of the

commodity is price elastic when there is expectation of a further future price reduction, this is so because with a decrease in the price of the commodity, the producers supply more of the commodity due to fear to sell their output at much lower price.

- **Availability of excess capacity:** The supply of a commodity is price elastic when they are firms are operating excess capacity, this is so because with increase in the price of a commodity, the producers easily increase production and supply of the commodity since the resources are not yet exhausted. On the other hand the supply of a commodity is price inelastic when the firms are operating at full capacity; this is so because with increase in the price of the commodity, the producers are not able to increase production and supply of the commodity.
- **Government policies of subsidization and taxation.** The supply of the commodity is price elastic when government subsidizes producers this is so because subsidies artificially reduce costs of production which encourages producers to produce and supply more of the commodity. On the other hand the supply of the commodity is price inelastic when the government highly taxes the producers, this is so because high taxes increases the costs of production which discourages producers thus reducing the supply of a given commodity

Factors responsible for elastic supply of a commodity

- Low costs of production.
- Shorter gestation period.
- Improved state of technology.
- Existence of durable commodities.
- Long-run period.
- Favourable natural factors especially in agricultural sector.
- Freedom of entry of firms in production.
- Favourable political climate.
- Expectation of further future price fall.
- Presence of excess capacity.
- Favourable government policy of subsidization.

Factors responsible for inelastic supply of a commodity

- High cost of production..
- Longer gestation period.
- Poor state of technology. .
- Presence of perishable commodities.
- Short-run period.
- Unfavourable natural factors especially in agricultural sector.
- Restricted entry of firms in production.
- Unfavourable political climate..
- Expectation of further future price rise. .

- Existence of full utilization of available resource.
- Unfavourable government policy of taxation.

IMPORTANCE OF PRICE ELASTICITY OF SUPPLY:

- Used in the devaluation policy. The supply of exports should be price elastic such that producers of exports should be able to increase production and supply more when there is increase in demand for exports,
- Used by government in formulating taxation policy. More tax revenue is be got by taxing goods that have got inelastic supply, this is so because imposition of taxes on such goods does not affect their supply greatly.
- Used to determine the incidence of a tax. A producer pays more of a tax when elasticity of supply is inelastic and pays less when the price elasticity of supply is elastic
- Used to determine the wage rate. Labour which has an inelastic supply earns a high wage because it is not easy to get such labour and labour with an elastic supply earns a low wage because it is easily acquired.

SUB-TOPIC7: PRICE MECHANISM AND PRICE CONTROL:

Price mechanism is a system in a free enterprise economy where by prices in the market are determined by the market forces of demand and supply and prices act as an automatic signal in the allocation of resources.

Under price mechanism there is **consumer sovereignty** where a consumer takes an upper hand in influencing the decision of the producers concerning what to produce, where to produce, for whom to produce , when to produce and how to produce. And there seems to be an invisible hand that allocates resources.

It follows that whenever a consumer buys a product, he is casting a vote in favour of that product. The more he buys the greater the producer is willing to supply on the market.

THE ROLE OF PRICE MECHANISM IN RESOURCE ALLOCATION IN AN ECONOMY:

1. **Provides automatic adjustment between supply and demand.** This is because an increase in demand for a commodity attracts more new firms into production of that commodity which results into increased supply thereby overcoming the shortage.
2. **Determines the type of technology to be used in production.** The producers employ the method of production which is efficient but at the same time affordable so as to produce high quality products in order to attract more buyers and hence make more profits.

3. **Promotes consumer sovereignty/determines for whom to produce;** producers supply goods for those who can afford to buy them or for those who are ready and willing to buy at high price so as to enable them maximise profits.
4. **It determines the income distribution .**Producers buy resources from resource owners and therefore income is distributed among the producers and resource owners. Those who own resources which are highly priced earn higher incomes as compared to those who own resources that are not highly priced.
5. **Determines where to produce /determines the location of the production unit.** Producers establish their business units where they can easily access customers who are ready to pay high prices for their products so as to enable them maximise profits.
6. **Determines when to produce.** Production always takes place at the time when consumers' demand dictates so and therefore they are ready to pay high prices. This is common with products that are demanded seasonally.
7. **Ensures production of better quality goods/ products.** This is so because of competition in production as a result of many producers engaging in production.
8. **Guides on what to produce.** Resources are allocated to production of those commodities that command high prices in the market so as to enable producers maximise profits.
9. **Ensures efficiency of firms.** Resources are usually allocated to the production of those commodities where minimum costs are incurred in order to fetch high prices and maximise profits.
10. **Guides in resource allocation (factor market).** Factors of production are attracted to areas where they are highly priced or highly paid.
11. **Promotes/encourages innovations and inventions.** This is intended to improve the method of production so as to minimise the costs of production in order to maximise profits.
12. **Provides variety of goods.** This is so because there are many producers in the market who produce a variety of goods so as to meet the different tastes and preference of the consumers in order to maximise profits, hence widening consumer's choice.

MERITS /ADVANTANTAGES/POSITIVE IMPLICATIONS OF PRICE MECHANISM:

1. It encourages the production of better quality goods; this is because production is competitive which leads to improvement in the quality of the products.
2. It promotes incentives for hard work leading to increased production. The profit motive encourages hard work, innovations and inventions hence increased productivity.
3. It encourages efficient allocations of resources because producers produce in response to consumer's demand therefore cases of over production and under production are avoided hence no wastage of resources.

4. It avails a wide variety of goods and services to consumers because there are many producers producing different commodities and this widens consumer's choice.
5. It encourages flexibility in production because producers adjust to changes in the market conditions basing on changing price for the commodity.
6. It is a cheap system to maintain/ it reduces the costs of administration because of limited government control since adjustments are automatic i.e. by forces of demand and supply.
7. It encourages arbitrage i.e. it facilitates regional distribution of goods which benefits producers since goods are transferred from areas of plenty where prices are low to areas of scarcity where prices are high.
8. Provides an incentive to economic growth. Higher prices and profits encourage large industrial establishments to spend huge sums of money on research, new and better techniques of production which leads to more production of goods and services hence accelerating economic growth.
9. It helps in the distribution of income; People owning large quantities of resources which are highly priced earn more income than those owning few resources.
10. It decentralises economic powers/ Promotes consumer sovereignty. This is so because individual households make their own economic decisions.
11. It leads to increased employment opportunities. This is due to increased economic activities/ increased investment as people strive to make more profits.

DEMERITS / DEFECTS/ SHORTCOMINGS/NEGATIVE IMPLICATIONS OF PRICE MECHANISM

1. It leads to uneven distribution of income or income inequality/disparity i.e. people with more productive resources earn more income than those with less productive resources.
2. It leads to the emergency of monopoly power and its associated evils. This happens when inefficient firms are outcompeted and driven out of the industry and the few that remain enjoy monopoly power which may lead to production of poor quality products, overcharging consumers etc.
3. It leads to consumer exploitation and this is due to ignorance of the consumers about the market conditions which results into charging them very high prices of goods and services.
4. It leads to emergency of unemployment. This happens when inefficient firms are outcompeted in business. Unemployment may also arise when firms adopt capital intensive techniques of production in bid to maximise profits by minimising costs. This results into technological unemployment.
5. It leads to divergence between the private and social benefits and costs. This is because price mechanism emphasizes the element of profit. Therefore producers aim at achieving their private

benefits while creating social costs for the society e.g., environment degradation resource pollution (water, air, and noise).

6. Inability to adjust to rapid structural changes in the economy. This is because it is forces of demand and supply and for producers to make any changes they wait for signals from the consumers.
7. It encourages wasteful competition and duplication of activities and this result into resource wastage as a result of intensive persuasive advertisements.
8. Makes the economy susceptible to economic instabilities such as price fluctuations. The producers tend to increase prices of goods and services with increased demand. This leads to inflationary tendencies in the economy.
9. Leads to distortion of consumer choices through persuasive advertisements.
10. **Inability to allocate resources to public and merit goods.** This is mainly because it would be impossible to charge them prices since free riders are not excluded in their consumption. In price mechanism producers aim at maximising profits so they tend to ignore such goods.
11. **Leads to misallocation of resources.** Price mechanism may not allocate resources to priority areas/ socially profitable ventures since it is guided by profit motive. This may lead to disappearances of cheap commodities on which the ordinary people survive.

REASONS FOR GOVERNMENT INTERFERENCE WITH THE PRICE MECHANISM:

- **To reduce unemployment;** this will be achieved by subsidising weak firms so that they continue producing so as to maintain employment opportunities in the country.
- **To ensure economic stability;** this will be achieved by using restrictive monetary policy which reduces the amount of money in circulation so as to stabilise prices.
- **To ensure production of essential goods and services;** this will be achieved by producing public goods which are ignored by the private sector.
- **To regulate production and provision of undesirable products (demerit goods);** this will be achieved by imposing high taxes on such goods to discourage their consumption.
- **To reduce on the level of income inequalities;** this will be achieved by using the progressive taxes where the rich are taxed more than the poor so as to redistribute incomes.
- **To control the growth and regulate the activities of monopolies;** this will be achieved by imposing taxes on their profits so as to redistribute incomes.
- **To reduce consumer ignorance due to market imperfections;** this will be done by encouraging formation of consumer associations to provide information about market conditions so that they are not cheated.
- **To reduce on social costs such as pollution and resource exhaustion that affect people's welfare;** this will be done by setting regulatory bodies/strict laws to regulate the actions of firms in resource exploitation and waste disposal.
- **To cater for the provision of public and merit goods;** this will be done by the government producing and providing them directly.
- **To counter rapid structural changes;** this will be achieved through proper economic planning as well as reserving funds which will effect the necessary adjustment in times of structural changes for example wars, war, famine, floods etc.

Factors that influence/affect/ determine price mechanism in resource allocation

1. **Availability of capital.** Adequate capital leads to better allocation of resources because the producers have the capital to purchase raw materials and hire skilled labour to increase production of goods and services. On the other hand, inadequate capital leads to inefficient allocation of resources because producers cannot purchase enough raw materials and hire skilled labour leading to low output levels.
2. **Level of labour skills,** low levels of skills of workers limit the operation of price mechanism because of low level of productivity and efficiency in production of goods and services lead to low productivity. On the other hand, high level of skills of workers promote price mechanism because of high level of efficiency among workers which lead to labour leads to high productivity.
3. **Level of entrepreneurial abilities.** High level of entrepreneurial abilities promotes initiative to undertake business ventures with aim of making profits thereby promoting efficient allocation of resources. On the other hand, low level of entrepreneurial abilities limits initiative to undertake business ventures in fear of risks involved thereby limiting price mechanism in the allocation of resources.
4. **Degree of monopoly power/number of firms.** High degree of monopoly power fails free operation of price mechanism as entry into economic activities by many firms is restricted to only one firm thereby limiting output levels. On the other hand, where there is no monopoly, entry into economic activities which enables by many firms to enter and increase production.
5. **Level of government interference.** High level of government interference in form of heavy taxation of the producers limits the operation of price mechanism because it raises the cost of production which discourages production of goods leading to shortages. On the other hand, low level of government interference promotes price mechanism because the government does not impose many restrictions to producers which lowers the cost of production and attract more producers,
6. **Level of ignorance/awareness of producers and consumers.** High level of consumer ignorance denies them the right to consume from cheapest sources which lowers the level of consumption while producer ignorance denies them the right to discover the cheapest sources of raw material which increases the cost of production and thus limiting the operation of price mechanism. On the other hand, high level of awareness about market conditions by the producers enable them to acquire raw materials at fair prices which lowers the cost of production while consumers are able to acquire commodities from the cheapest sources thus promoting price mechanism.
7. **Ability to forecast future trends in demand.** Price mechanism works better where producers are to predict changes in the conditions of demand and supply in future to enable them supply and satisfy the demand. On the other hand price mechanism fails to work effectively where producers cannot predict changes in demand because they either over supply or under supply which leads to wastage of resources.

8. **The degree of rationality of consumers and producers.** Price mechanism fails where producers and consumers are irrational because producers fail to minimise costs by buying raw materials expensively leading to low profits while consumers fail to maximise utility by buying goods from expensive sources. On the other hand price mechanism works better where producers and consumers are rational because the producers are able to acquire raw materials at fair prices which increase production while consumers purchase more goods since they purchase from the cheapest source.
9. **The state of technology.** Poor state of technology limits the operation of price mechanism because of the high cost of production which discourages producers from employing producers more resources. On the other hand improved state of technology promotes effective operation of price mechanism because it lowers the cost of production which enables producers to employ more resources and increase production.
10. **The state of infrastructure.** Poor infrastructure in the form of poor roads limits the operation of price mechanism because it increases the costs of distributing goods(arbitrage) leading to shortages in some areas. On the other hand, for example better roads promote the operation of price mechanism because it lowers the costs of distribution of goods to different areas which promotes even distribution of goods.
11. **The degree mobility of factors of production.** High degree of factor mobility especially labour promotes effective operation of price mechanism because factors of production easily move to where they are needed which minimises shortage of labour and ensure continuous production of goods. On the other hand, high degree of factor immobility limits the operation of price mechanism because factors of production easily to where they are needed which leads to shortage of labour in some places which brings production to standstill.
12. **Political climate;** a stable political climate leads to effective operation of the price mechanism because it ensures continuous production of goods and effective distribution goods since producers are assured of the safety of their lives and property. On the other hand political instability limits the operation of price mechanism because production and distribution of goods is interfered with since producers are scared of losing their lives and property.

LIMITATIONS OF PRICE MECHANISM IN ALLOCATION OF RESOURCES

- Government interference/ regulation.
- Existence of monopoly. .
- Immobility of factors of production. .
- Inability to forecast future trends
- Limited entrepreneurship.
- Irrationality of producers and consumers. .
- Limited capital.
- Limited skilled labour.
- Poor infrastructures.
- Ignorance about market conditions by producers and consumers.
- Poor state of technology.

- Political instability.

FACTORS THAT PROMOTE PRICE MECHANISM IN THE ALLOCATION OF RESOURCES:

- Limited government interference
- Existence of many producers
- High degree of factor mobility
- High degree of ability to forecast future trend
- High level of rationality among producers and consumers
- Improved state of technology
- Improved state of infrastructure
- High degree of mobility of factors of production
- Political stability
- Adequate supply of capital
- High labour skills
- High level of entrepreneurial skills/ability
- High degree of awareness about market conditions by producers and consumers.

MEASURES THAT CAN BE TAKEN TO CONTROL/REDUCE THE DEMERITS OF PRICE MECHANISM:

1. **Introduce progressive taxation;** this will ensure that the rich are taxed more than the poor to reduce income inequalities.
2. **Provide incentives to protect/prop up weak firms;** this will lower their cost of production and thus enable them to continue producing in order to maintain employment opportunities.
3. **Issue licences;** this will help to regulate the activities of the investors by restricting the number of licences issued so as to control over exploitation of resources.
4. **Encourage formation of consumers' protection organisations/Associations;** this will create awareness among the consumers on issues concerning the prices and the quality of goods on the market so that they are not exploited.
5. **Introduce price controls to ensure stable prices;** this can be achieved through setting maximum price to protect consumers and minimum price to protect producers against exploitation.
6. **Put in place anti-monopoly laws;** this will guard against emergence of monopoly firms so as to minimise their evils.

7. **Provide public goods and merit goods;** this will be achieved by the government allocating resources to produce such goods since they are ignored by the private investors.
8. **Undertake proper planning;** this will be done in order to effect the necessary adjustments in order to counter structural changes.
9. **Enact laws/set up regulatory bodies;** this will regulate economic activities by setting guidelines on how to use the resources thus minimise environmental degradation e.g. NEMA and NFA.
10. **Introduce rationing;** this will involve direct action by the public authorities apportioning scarce supplies to all households on a regular basis during periods of scarcity to ensure that all people get access to essential goods and services.

PRICE CONTROL/PRICE LEGISLATION: This refers to the situation where the government fixes the prices of commodities either a maximum price to protect consumers or a minimum price to protect producers. Such a price which is fixed is referred to as an administered price.

Minimum price Legislation; This is the setting/fixing of prices of commodities by the government above equilibrium price below which it becomes illegal to buy the commodity. It protects producers.

NB: Price support; This is where the government buys the surplus output in the market arising out of fixing the minimum price in order to avoid discouraging producers.

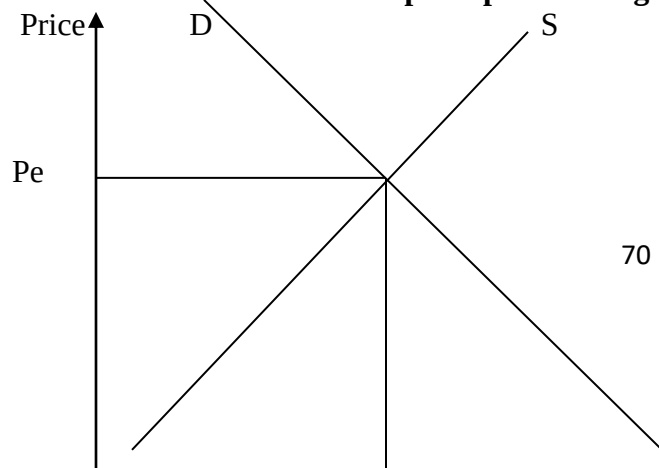
Maximum Price Legislation; This is the setting/ fixing of prices of commodities by the government below the equilibrium price above which it becomes illegal to sell the commodity. It protects consumers.

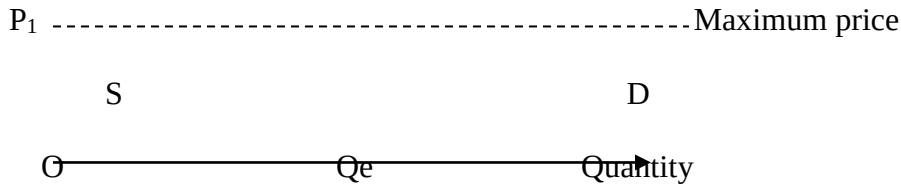
MAXIMUM PRICE/ PRICE CEILING; this is the legal price set by the government below the equilibrium price above which it is illegal to sell the commodity.

The major aim of fixing the maximum price is to protect consumers from being exploited by profit hungry traders.

The maximum price is also referred to as price ceiling because it is the highest price above which nobody should sell the commodity.

An illustration of Maximum price/price ceiling





In the diagram above, the maximum price is set at OP_1 below the equilibrium price OP_e .

REASONS FOR SETTING A MAXIMUM PRICE/PRICE CEILING;

1. To protect consumers from being exploited by the business men through overcharging. It is illegal to sell above it.
2. To encourage the consumption of a particular commodity, most especially merit goods, because the price is made low enough for consumer
3. To maintain price stability through controlling inflation; this is because sellers are not supposed to charge above the maximum price.
4. To control monopoly tendencies; this is because monopolists are not supposed to charge above the administered price.

MERITS / POSITIVE IMPLICATIONS OF SETTING A MAXIMUM PRICE:

1. It protects the consumers from being over charged by the producers
2. It helps to make commodities available to all income groups of people in the economy.
3. It enables to maintain price stability in an economy since the sellers are not supposed to sell the commodities above the maximum price.
4. It helps to control monopolistic tendencies and its associated evils such as overcharging, since the monopolist is not supposed to sell above the maximum price.

DEMERITS/ NEGATIVE EFFECTS OF SETTING A MAXIMUM PRICE:

1. It leads to artificial shortage of goods; this because the price is too low to encourage supply and yet consumers are willing to buy
2. It encourages trade malpractices such as smuggling, hoarding and black marketing; this is because the business people want to sell at high prices.

NB: A black market is a situation where producers / sellers sell goods illegally at a price higher than the price fixed by the government.

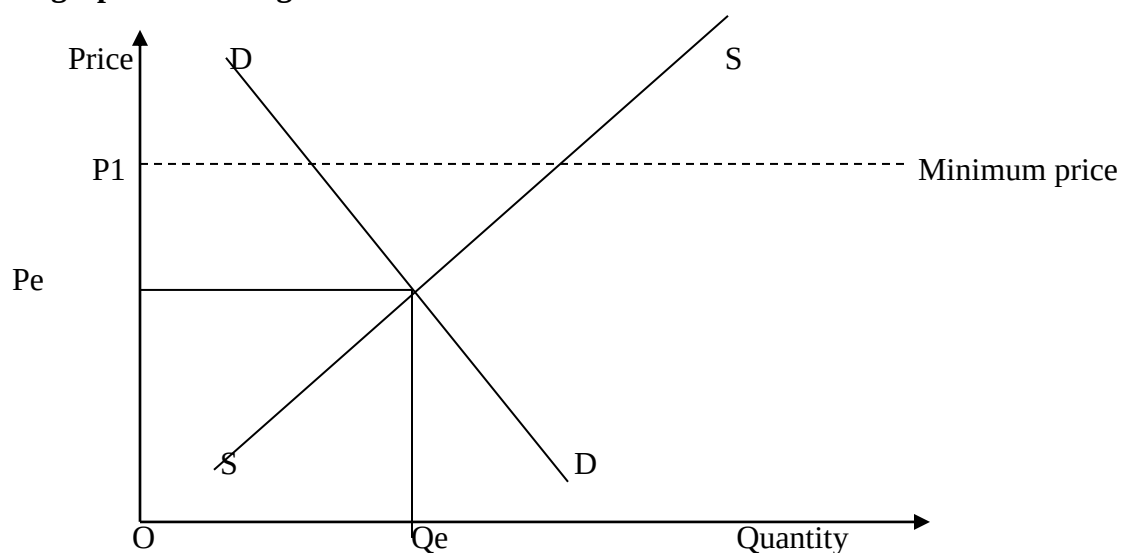
3. It leads to underutilisation of resources; because producers produce at excess capacity since they are discouraged by the low price.
4. It reduces incentives for private entrepreneurs; because it lowers the profit levels which discourage people from setting up firms.
5. It leads to high government expenditure; because of the high administrative costs incurred by the government to employ scouts/ enforcement officers to ensure that goods are sold at the administered prices.
6. **It leads to unemployment;** this is due to reduced levels of investment since the maximum price discourages the investors.
7. It leads to misallocation/inefficient allocation of resources; this is because the government sets the prices and ignores the market forces of demand and supply.

MINIMUM PRICE/PRICE FLOOR; this is the price set by the government above the equilibrium price, below which it becomes illegal to buy the commodity.

The price is set in order to protect the producers from being exploited by the buyers.

It is also called the price floor because it is the lowest price below which a commodity cannot be traded.

A graph illustrating Minimum Price/ Price floor



A minimum price OP1 is set above the equilibrium price OPe

REASONS FOR SETTING A MINIMUM PRICE:

1. To protect producers from being exploited by the buyers who normally offer low prices for their products; this is because it is illegal to buy the commodity below the administered price.
2. To control price fluctuations of agricultural products that are affected by changing climatic conditions; this because the price is set by the government and cannot change unless the government decides so.
3. To encourage mass production of goods/ to accelerate the rate of economic growth; this is because the price is high enough to encourage production.
4. To offset an economic recession/ depression; this price is high enough to encourage production which increases incomes of the people and thus be able to buy goods.
5. **To encourage creativity and innovativeness in production**; this is because high price motivate them to innovate and increase production
6. To minimise the exploitation of labour; this is through setting a minimum wage which makes it illegal to pay a worker below it.

MERITS/POSITIVE IMPLICATIONS OF SETTING A MINIMUM PRICE:

1. Protects producers from being exploited by consumers
2. Enables producers to realise stable incomes.
3. Leads to increased production of goods and services. .
4. Helps an economy out of an economic depression / recession.
5. Helps to establish industrial peace.
6. It minimises exploitation of labour.

DEMERITS/ NEGATIVE IMPLICATIONS OF SETTING A MINIMUM PRICE:

1. Leads to unmanageable surpluses; because the minimum price motivates the producers to produce more goods and yet consumers are discouraged to buy.
2. It leads reduction in social welfare; this is because the minimum price is set above the equilibrium price and therefore it becomes expensive for the buyers to purchase from the producers.

3. **It leads to high administrative costs.** This is because the government employs scouts to monitor the set price in order to ensure that goods are sold at the fixed price. At the same time, the government is forced to buy the surplus output through price support system.
4. **It increases the cost of production;** this is because of the high cost of labour due to a minimum wage.
5. **It leads to technological unemployment;** this because entrepreneurs adopt capital intensive technology to minimise the high cost of labour due to a minimum wage.
6. It leads to misallocation/inefficient allocation of resources; this is because the government sets the prices and ignores the market forces of demand and supply.

Reasons for legislating prices by the government include;

- To protect consumers against exploitation from greedy businessmen.
- To control monopoly power
- To control inflation/ ensure price stability
- To protect producers from being exploited by the consumers
- To increase output hence accelerate economic growth.
- To make commodities available to all groups of people
- To help establish industrial peace
- To stabilise producers' incomes
- To reduce income inequality
- To help an economy offset an economic depression

Demerits of price legislation;

- Leads to unmanageable surpluses
- Leads to unemployment due to reduced investment
- It reduces incentives for private entrepreneurs
- Leads to shortages in supply due to increased demand
- Leads to inefficient allocation of resources due to distortion of price mechanism
- It is expensive for the government to enforce because of high administrative costs.
- It encourages trade malpractices, e.g. smuggling
- It leads to increased costs of production
- It leads to reduction in social welfare due to high cost of living
- Leads to underutilisation of resources due to production at excess capacity.

SUB-TOPIC 8: PRICE FLUCTUATIONS:

Price fluctuations refer to unstable prices of goods over a given period of time and in this case it refers to the state of upward and downward movement of prices especially of agricultural products. There is greater oscillations/fluctuations in prices of agricultural goods as compared to prices of manufactured goods.

THE CAUSES OF AGRICULTURAL PRICE FLUCTUATIONS:

1. **The long gestation period.** Agricultural commodities have long gestation period and their supply cannot be increased in the short run which forces prices to **rise** due to scarcity. However in the long run after harvesting the supply increases which forces the prices to **fall** due to excess supply on the market.
2. **Agricultural products are bulky hence difficult to transport.** When there is surplus output in an area it becomes difficult to transport the excess supply to other areas which leads to a fall in price, however during period of scarcity it is hard to transport products to the area which leads to an increase in price.
3. **Heavy dependence on natural factors which affect the level of output;** Unfavourable natural factors such as prolonged drought, floods, pests and diseases reduce the supply of agricultural products which forces the prices to **rise/increase**. On the other hand favourable natural factors such as reliable rainfall, fertile soil increase the supply of agricultural products which leads to a fall/**decrease** in prices of such products.
4. **Agricultural products are perishable.** Agricultural products are highly perishable and they cannot be stored for a long time. This forces the producers to sell them quickly which **reduces** prices due to excess supply. However after most of the produce is sold, the supply reduces which forces the prices to **rise/increase** due to scarcity.
5. **Divergence between planned and actual output.** When actual output is greater than planned output due to favourable natural factors, it leads to excess supply of agricultural products on the market thus, leading to a **fall/decrease** in their prices. On the other hand when the actual output is less than the planned output due to unfavourable natural factors, it leads to decrease in the supply of agricultural output which leads to a **rise/increase** in their prices.
6. **Many producers hence planning is difficult/Poor planning by the farmers.** When farmers fetch high prices in one season, they plan to produce more in the following season which increases supply and thus forces the prices to **fall**. On the other hand when the prices are low in one season farmers plan to produce less in the following season this forces the prices to **rise** because of the shortages/ scarcity.
7. **Weak bargaining position on the world market/External determination of prices of agricultural products;** the major buyers of agricultural products dictate the prices of agricultural products, when supply of agricultural products increases, they offer low prices in the market, this forces the middlemen to **reduce** the prices paid to the local farmers. On

the other hand when supply reduces the buyers offer high prices because they compete for the little output and this enables the middlemen to **increase** the prices paid to the local farmers.

8. **Low price elasticity of demand for agricultural products;** when quantity supplied increases the sellers quickly **reduce** the price to clear the market, and when supply reduces they quickly **increase** the price since consumers still buy even at higher prices.
9. **Low income elasticity of demand for agricultural products;** farmers increase production as the incomes of the buyers increase hoping to sell more and earn high profits, however the buyers maintain the quantity bought which leaves the farmers with excess output and thus **reduction** in price and sellers **increase** price as output reduces since the consumers are able to buy even at higher prices.
10. **Poor surplus disposal system/machinery e.g. poor infrastructure;** the poor transport system makes it hard to transport excess output to regions of scarcity therefore prices **fall** in areas of surplus in order to minimise loss and when supply reduces prices rise since little was stored and it is also difficult to increase supply from other areas.
11. **Change in cost of production.** Increase in cost of production compels farmers to reduce production which reduces output leading to an increase in prices while a decrease in cost of production attracts many farmers to produce which increases production leading to a fall in prices
12. **Substitution of agricultural products with synthetic products.** when the demand for synthetic products increases the demand for agricultural products reduces leading to a fall in their prices on the other hand a decrease in demand for synthetic products ,the demand for agricultural products increases leading to an increase their prices.
13. **Innovation of raw material saving techniques.** Increased use of raw material saving technology reduces the demand for agricultural products leading to a fall in their prices while reduced use of raw material saving technology the demand for agricultural products increases leading to a rise in their prices.
14. **Weak commodity agreements.** These fail to regulate supply and as a result prices reduce when members over supply and prices increase when supply reduces.

THE EFFECTS OF AGRICULTURAL PRICES FLUCTUATIONS:

1. **It leads to unstable export earnings.** In some seasons an **increase** in export prices increases exports earnings while in some seasons a **decrease** in export prices leads to a **fall** in export earnings.
2. **Leads to fluctuations exchange rates;** an **increase** export prices leads to an increase foreign exchange inflow which results to a decline in the exchange rate and a **fall** in export prices lead to reduction in foreign exchange inflow which leads to a rise in the exchange rate.
3. **Leads to unstable government revenue;** an **increase** prices of agricultural products widens the tax base which increases government revenue since earnings of the farmers'

increase and a **fall** in prices of agricultural products reduces the tax base which leads a decrease in government revenue due to the fall in the farmers' incomes

4. **It leads to fluctuations in the balance of payment position;** an **increase** export prices results into improvements of the balance of payment position due to an increase in foreign exchange inflow and a **fall** in export prices worsens of the balance of payment position due to a fall in foreign exchange earnings.
5. **It leads to fluctuations incomes of the farmers/ agricultural producers;** an **increase** in the prices of agricultural products leads to an increase in the incomes of the farmers' and a **fall in** prices of agricultural products leads to a fall in the farmers' incomes.
6. **Leads to unstable terms of trade;** increase prices of agricultural exports leads improvement in the terms of trade and a fall in prices of agricultural exports leads to decline in the terms of trade.
7. **It worsens the problem of income inequalities;** a fall in prices of agricultural products makes the farmers to earn much less compared to people in other sectors of the economy which worsens income inequality.
8. **Investment in agriculture becomes uncertain;** this is because farmers earn unstable incomes which make lose interest in farming and thus some abandon agricultural production.
9. **It leads to unemployment;** a fall in prices of agricultural products reduces investment in the sector and thus lowers employment opportunities.
10. **It encourages rural urban migration and its associated evils;** the fall in prices of agricultural products frustrates many farmers which pushes them to urban areas in search of better employment opportunities.
11. **It makes projected planning difficult;** this is because prices change from time to time hence making earnings uncertain which necessitates redrawing of plans from time to time.

MEASURES THAT CAN ADOPTED/TAKEN TO STABILISE PRICES OF AGRICULTURAL PRODUCTS IN AN ECONOMY:

1. **Undertake agricultural diversification;** this will encourage the farmers to undertake several economic activities within the sector, this will avoid over production of one commodity thus minimizing surplus.
2. **Improve on the transport system;** this will ease transportation of agricultural products from areas of plenty to areas of scarcity which will stabilise prices in all the areas.
3. **Promote agro-based industries;** this will provide a stable market for agricultural products which help to avoid excess supply and thus ensure stable prices or add value to the products to prolong their shelf life.
4. **Strengthen/Join international commodity agreements;** this will improve the bargaining power of the producers' agricultural products in the international markets which will help to stabilise prices of such products. They also regulate supply to avoid over or under production.
5. **Modernise agriculture for example encourage irrigation;** this will reduce dependence on nature which will ensure continuous production throughout and thus stabilise supply and prices of agricultural products

6. **Undertake market expansion and diversification;** this will create wide market for the agricultural products which will help to avoid flooding the market, hence ensure stable prices.
7. **Encourage contract farming;** this will encourage the farmers to sign contracts with the consumers about the quantity and price before production takes place which will ensure stable prices. E.g. a poultry farmer can sign a contract with a hotel manager to supply chicken and eggs at an agreed price before production takes place.
8. **Encourage cooperative societies;** these will market the produce for the members and negotiate for fair prices for such products as well as regulating supply to avoid over or under production.
9. **Improve storage facilities;** this will regulate supply by storing excess output during period bumper harvest and releasing it during the period of scarcity which will help to avoid over flooding in the market and thus stabilise prices of such products.
10. **Use/Operate buffer stock;** this will involve buying surplus output by marketing boards during bumper/plenty harvest and sold in periods of scarcity which will help to stabilise prices.
11. **Use stabilisation fund policy;** this will involve buying the surplus output by the government, keeps it and sells it later at a profit which accumulate a fund out of which it will pay farmers as compensation for the reduction in incomes and thus help to stabilise their incomes
12. **Set minimum price for agricultural products;** this will ensure that agricultural commodities are sold at the fixed price and cannot be changed unless the government decides so.
13. **Adopt a strict quota system;** this will regulate the supply of agricultural products so as to avoid flooding the market and thus stabilise prices
14. **Subsidise farmers.** This will ensure stable costs of production, which will ensure stable output levels leading to stable prices.